Cambridge International AS Level Physics (9702)

305 Extreme-Difficulty Practice Questions

Detailed Answers & Worked Solutions

Pack 2 | AS syllabus 2025-2027 | Target series: Oct / Nov 2026 | Topics 1–11

Full method, working and mark-by-mark reasoning for every question

Take g = 9.81 m s^{-2} and c = 3.00 × 10^8 m s^{-1} unless stated otherwise.

SECTION A — Structured (theory) answers [Q1–Q140]

1 Physical quantities and units

Q1. The drag force on a car is modelled by $F = ½C\rho Av^2$, where ρ is the air density, A the frontal area and v the speed. **[7]**

1. Use SI base units to show that the constant C is dimensionless.
2. A car of frontal area 2.1 m^2 travelling at 25 m s^{-1} in air of density 1.2 kg m^{-3} experiences a drag force of 560 N. Calculate C.

Answer. **(a)** ρAv^2 has base units (kg m^{-3})(m^2)(m^2 s^{-2}) = kg m s^{-2} = N, the unit of F. So ½C must be dimensionless ⇒ **C is dimensionless**.
(b) C = 2F/(ρAv^2) = 2×560/(1.2×2.1×25^2) = 1120/1575 = **0.71**.

Q2. Estimates are an important physics skill. **[6]**

1. Estimate the number of times a human heart beats in a lifetime, showing your assumptions.
2. Given that a heart pumps about 70 cm^3 of blood per beat, estimate the total volume of blood pumped in a lifetime.

Answer. **(a)** Assume ~70 beats per minute and a lifetime of ~75 years. Beats = 70 × 60 × 24 × 365 × 75 ≈ **3 × 10^9 beats** (any answer 2–4 × 10^9 acceptable).
(b) Volume ≈ 3×10^9 × 70×10^{-6} m^3 = **2 × 10^5 m^3** (about 2×10^8 litres).

Q3. Express each of the following derived units in terms of SI base units only. **[6]**

1. the pascal (Pa)
2. the watt (W)
3. the volt (V)

Answer. **(a)** Pa = N m^{-2} = kg m s^{-2} · m^{-2} = **kg m^{-1} s^{-2}**.
(b) W = J s^{-1} = kg m^2 s^{-2} · s^{-1} = **kg m^2 s^{-3}**.
(c) V = W A^{-1} = kg m^2 s^{-3} · A^{-1} = **kg m^2 s^{-3} A^{-1}**.

Q4. A student times a simple pendulum, for which $T = 2\pi\sqrt{L/g}$. They measure the length L = (0.994 ± 0.002) m and the time for 20 complete oscillations as (39.8 ± 0.4) s. **[8]**

1. Calculate a value for g.
2. Determine the percentage uncertainty in g and hence its absolute uncertainty.

> **Answer.** T = 39.8/20 = 1.99 s.
> **(a)** $g = 4\pi^2L/T^2 = 4\pi^2(0.994)/1.99^2 = 39.24/3.960 =$ **9.91 m s^{-2}**.
>
> **(b)** %L = 0.002/0.994 = 0.20%; %T = 0.4/39.8 = 1.005%, and T is squared ⇒ 2.01%. %g = 0.20 + 2.01 = **2.2%**. Δg = 0.022 × 9.91 = 0.22 ⇒ g = **(9.9 ± 0.2) m s^{-2}**.

Q5. In an experiment the potential difference across a resistor is (6.0 ± 0.1) V and the current is (0.25 ± 0.01) A. **[5]**

1. Calculate the resistance.
2. Determine the absolute uncertainty in the resistance and state the result appropriately.

> **Answer. (a)** R = V/I = 6.0/0.25 = **24 Ω**.
>
> **(b)** %R = %V + %I = 0.1/6.0 + 0.01/0.25 = 1.67% + 4.0% = 5.67%. ΔR = 0.0567 × 24 = 1.4 ⇒ R = **(24 ± 1) Ω**.

Q6. A 12 N weight hangs from a light string. A horizontal force H is applied to the weight so that the string makes an angle of 25° with the vertical and the weight is in equilibrium. **[6]**

1. Draw a free-body diagram and resolve forces to find the tension in the string.
2. Find the horizontal force H.

> **Answer.** Three forces: weight 12 N (down), tension T (along string), horizontal push H.
>
> **(a)** Vertical: T cos25° = 12 ⇒ T = 12/0.906 = **13.2 N**.
> **(b)** Horizontal: H = T sin25° = 13.2 × 0.423 = **5.6 N**.

Q7. Two forces of magnitude 5.0 N and 8.0 N act at a point with an angle of 50° between them. **[6]**

1. Calculate the magnitude of their resultant.
2. Find the angle between the resultant and the 8.0 N force.

> **Answer. (a)** $R^2 = 5.0^2 + 8.0^2 + 2(5.0)(8.0)\cos 50° = 25 + 64 + 80(0.643) = 140.4$ ⇒ R = **11.9 N**.
>
> **(b)** tanα = 5.0 sin50°/(8.0 + 5.0 cos50°) = 3.83/11.21 = 0.342 ⇒ α = **18.9°** from the 8.0 N force.

Q8. An analogue ammeter reads 0.20 A when no current passes through it. A student then uses it to record four readings of a steady current: 2.05, 2.07, 2.06, 2.04 A. **[6]**

1. Name the type of error indicated by the reading of 0.20 A with no current, and state the corrected mean current.
2. Comment on the precision of the four readings, and explain why high precision does not guarantee an accurate result here.

> **Answer.** **(a)** A non-zero reading with no current is a **zero (systematic) error**. Mean reading = 2.055 A; corrected current = 2.055 - 0.20 = **1.86 A**.
> **(b)** The readings agree to within ±0.02 A, so they are **precise** (small random scatter). But precision only reflects repeatability; the constant 0.20 A zero error shifts every reading the same way, so the uncorrected values are systematically too high — precise but inaccurate until corrected.

Q9. The speed v of a transverse wave on a stretched string is thought to depend on the tension T in the string and its mass per unit length μ (base units kg m^{-1}), such that $v = kT^{a}\mu^{b}$ with k a dimensionless constant. **[7]**

1. Use base units to determine a and b.
2. Write down the resulting relationship.

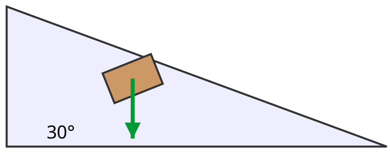
> **Answer.** Base units: v = m s^{-1}; T = kg m s^{-2}; μ = kg m^{-1}.
>
> **(a)** m s^{-1} = $(\text{kg m s}^{-2})^{a}(\text{kg m}^{-1})^{b}$ = $\text{kg}^{a+b}\ \text{m}^{a-b}\ \text{s}^{-2a}$. s: -2a = -1 $\Rightarrow$ a = ½. kg: a + b = 0 $\Rightarrow$ b = -½. (check m: a - b = 1 □). So **a = ½, b = -½**.
>
> **(b)** $\mathbf{v = k\sqrt{T/\mu}}$.

Q10. Estimate the average pressure exerted on the floor by a person of mass 70 kg standing still. State clearly the estimate you make for the contact area and give your answer to an appropriate number of significant figures. **[4]**

> **Answer.** Weight = 70 × 9.81 = 687 N. Estimate the contact area of two shoes as about 0.02 m^2 ($\approx$ 200 cm^2). p = F/A = 687/0.02 $\approx$ **3×10^4 Pa** (any 2–5 × 10^4 Pa with a sensible area is acceptable).

Q11. A block of weight 45 N rests on a frictionless slope inclined at 30° to the horizontal. **[4]**

1. Resolve the weight into components parallel and perpendicular to the slope.
2. State the magnitude of the force, directed up the slope, needed to hold the block at rest.



Answer. **(a)** Parallel to slope: 45 sin30° = **22.5 N**. Perpendicular: 45 cos30° = **39.0 N**.
(b) To hold the block, the up-slope force must balance the parallel component: **22.5 N**.

Q12. A ball moving due east at 5.0 m s^{-1} is struck and afterwards moves due north at 5.0 m s^{-1}. **[5]**

1. By drawing a vector diagram, find the magnitude of the change in velocity.
2. State the direction of this change.

Answer. Change in velocity = v_{final} - $v_{initial}$ = (north 5.0) + (west 5.0) (subtracting an eastward vector adds a westward one).

(a) $|\Delta v| = \sqrt{5.0^2 + 5.0^2}$ = **7.1 m s^{-1}**.
(b) Directed **45° west of north** (i.e. towards the north-west).

Q13. A steel ball bearing has mass (28.2 ± 0.1) g and diameter (19.0 ± 0.1) mm. **[7]**

1. Calculate the density of the steel.
2. Determine the percentage uncertainty in the density.

Answer. $V = (\pi/6)d^3 = (\pi/6)(0.0190)^3 = 3.59 \times 10^{-6}$ m^3.
(a) $\rho = m/V = 0.0282/3.59\times10^{-6}$ = **7.9×10^3 kg m^{-3}**.

(b) %m = 0.1/28.2 = 0.35%; %d = 0.1/19.0 = 0.53%, and d is cubed $\Rightarrow$ 1.58%. %ρ = 0.35 + 1.58 = **1.9%**.

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2 Kinematics

Q14. A football is kicked from level ground with a speed of 22 m s^{-1} at 40° above the horizontal. Air resistance is negligible. **[8]**

1. Calculate the maximum height reached.
2. Calculate the time of flight.
3. Calculate the horizontal range.

Answer. u_x = 22 cos40° = 16.9; u_y = 22 sin40° = 14.1 m s^{-1}.
(a) H = $u_y^2/(2g)$ = 14.1^2/19.62 = **10.2 m**.
(b) t = $2u_y/g$ = 2×14.1/9.81 = **2.88 s**.
(c) R = $u_x t$ = 16.9 × 2.88 = **48.6 m**.

Q15. A car starts from rest and accelerates uniformly at 2.0 m s^{-2} for 6.0 s. It then travels at constant speed for 20 s, before decelerating uniformly at 3.0 m s^{-2} to rest. **[9]**

1. Find the maximum speed reached.
2. Find the total distance travelled.
3. Find the average speed for the whole journey.

Answer. **(a)** v = u + at = 0 + 2.0×6.0 = **12 m s^{-1}**.
(b) Phase 1: s = ½(2.0)$(6.0)^2$ = 36 m. Phase 2: s = 12×20 = 240 m. Phase 3: time = 12/3.0 = 4.0 s, s = ½(12)(4.0) = 24 m. Total = 36 + 240 + 24 = **300 m**.
(c) Total time = 6.0 + 20 + 4.0 = 30 s. Average speed = 300/30 = **10 m s^{-1}**.

Q16. A ball is thrown horizontally at 15 m s^{-1} from the top of a vertical cliff of height 45 m. Air resistance is negligible. **[8]**

1. Calculate the time taken to reach the sea.
2. Calculate the horizontal distance travelled.
3. Calculate the magnitude and direction of the velocity as it hits the sea.

Answer. **(a)** 45 = ½gt^2 ⇒ t = $\sqrt{90/9.81}$ = **3.03 s**.
(b) Range = 15 × 3.03 = **45.4 m**.
(c) v_y = gt = 9.81 × 3.03 = 29.7 m s^{-1}. Speed = $\sqrt{15^2 + 29.7^2}$ = **33.3 m s^{-1}** at $\tan^{-1}$(29.7/15) = **63° below the horizontal**.

Q17. In a reaction-time test, a ruler is released from rest and a student catches it after it has fallen 0.18 m. **[4]**

1. Calculate the student's reaction time.
2. State one assumption made.

> **Answer. (a)** $s = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{2s/g} = \sqrt{0.36/9.81} =$ **0.19 s**.
> **(b)** Air resistance is negligible (or the ruler is released cleanly and caught the instant the student reacts).

Q18. A ball is thrown vertically upward from ground level with a speed of 25 m s^{-1}. Air resistance is negligible. **[6]**

1. Find the two times at which the ball is at a height of 20 m.
2. Explain physically why there are two such times.

> **Answer. (a)** $20 = 25t - \frac{1}{2}(9.81)t^2 \Rightarrow 4.905t^2 - 25t + 20 = 0$. $t = [25 \pm \sqrt{625 - 392.4}]/9.81 = [25 \pm 15.25]/9.81$. t = **0.99 s** and **4.10 s**.
> **(b)** The ball passes 20 m once on the way up and again on the way down; between these times it rises to its maximum height and returns.

Q19. The velocity–time graph of a cyclist over 40 s consists of three straight sections: a rise from 0 to 8.0 m s^{-1} in the first 10 s, a constant 8.0 m s^{-1} for the next 20 s, and a fall from 8.0 m s^{-1} to 2.0 m s^{-1} over the final 10 s. **[7]**

1. Describe the motion in each of the three sections.
2. Calculate the total distance travelled by finding the area under the graph.

> **Answer. (a)** 0–10 s: uniform acceleration (0.80 m s^{-2}); 10–30 s: constant velocity (zero acceleration); 30–40 s: uniform deceleration (0.60 m s^{-2}).
> **(b)** Area = ½(10)(8.0) + (20)(8.0) + ½(8.0 + 2.0)(10) = 40 + 160 + 50 = **250 m**.

Q20. Describe an experiment to determine the acceleration of free fall *g* using an electronically timed falling steel ball. Your answer should include: **[6]**

1. the measurements taken and the apparatus used;
2. how a graph is used to obtain g and reduce the effect of random error.

> **Answer. (a)** A steel ball is released from an electromagnet; a timer starts as the ball is released and stops when it strikes a trap-door/light gate. Measure the fall distance s (metre rule) for several values and the corresponding fall time t (electronic timer). Repeat each and average.
> **(b)** Since $s = \frac{1}{2}gt^2$, plot s against t^2: a straight line through the origin of gradient ½g, so g = 2 × gradient. Using many points and a best-fit line averages out random timing errors and reveals any systematic offset (e.g. a non-zero intercept).

Q21. A stone falls freely past two marks P and Q on a wall. It passes P and then Q, a distance 1.50 m lower, 0.100 s later. **[8]**

1. Find the speed of the stone at P.
2. Find the height above P from which the stone was dropped from rest.

> **Answer.** **(a)** Between P and Q: $s = ut + ½gt^2 \Rightarrow 1.50 = u(0.100) + ½(9.81)(0.100)^2 = 0.100u + 0.049 \Rightarrow u = 1.451/0.100 =$ **14.5 m s^{-1}** (speed at P).
>
> **(b)** From rest to P: $u^2 = 2gh \Rightarrow h = 14.5^2/(2×9.81) = 210/19.62 =$ **10.7 m**.

Q22. A projectile is launched from ground level at 20 m s^{-1} at 55° above the horizontal towards a vertical wall of height 4.0 m situated 6.0 m away. **[7]**

1. Find the time to reach the wall horizontally.
2. Determine whether the projectile clears the top of the wall, and by how much.

> **Answer.** u_x = 20 cos55° = 11.5; u_y = 20 sin55° = 16.4 m s^{-1}.
> **(a)** t = 6.0/11.5 = **0.52 s**.
> **(b)** $y = u_y t - ½gt^2 = 16.4(0.523) - 4.905(0.523)^2 = 8.57 - 1.34 = 7.2$ m. Since 7.2 m > 4.0 m, it **clears the wall by about 3.2 m**.

Q23. A ball A is released from rest at the top of a tower of height 30 m. At the same instant a ball B is thrown vertically upward from the base of the tower with a speed of 15 m s^{-1}. Air resistance is negligible. **[7]**

1. Find the time at which the two balls are at the same height.
2. Find that height above the ground.

> **Answer.** Measure heights from the ground. A: $y_A = 30 - ½gt^2$. B: $y_B = 15t - ½gt^2$.
>
> **(a)** Equal: $30 - ½gt^2 = 15t - ½gt^2 \Rightarrow 30 = 15t \Rightarrow t =$ **2.0 s** (the $-½gt^2$ terms cancel).
> **(b)** $y = 15(2.0) - ½(9.81)(2.0)^2 = 30 - 19.6 =$ **10.4 m**.

Q24. A body starts from rest and moves with uniform acceleration 4.0 m s^{-2} in a straight line. **[5]**

1. Show that the distance travelled during the 5th second of its motion is 18 m.
2. Explain why this exceeds the distance travelled in the 1st second.

> **Answer.** **(a)** Distance in nth second = $s_n - s_{n-1} = ½a(2n - 1)$. For n = 5: ½(4.0)(9) = **18 m**.
> **(b)** The body is continually speeding up, so it covers more ground in each successive second; in the 1st second (n = 1) the distance is only ½(4.0)(1) = 2.0 m.

Q25. A ball is thrown vertically upward and returns to the thrower's hand. Air resistance is negligible. **[6]**

1. Sketch the acceleration–time graph for the whole motion.
2. Sketch the velocity–time graph, and mark the point corresponding to maximum height.
3. State what feature of the velocity–time graph represents the displacement.

> **Answer.** **(a)** A horizontal line at -9.81 m s^{-2} for the whole time (acceleration is constant, downward, even at the top).
> **(b)** A straight line of constant negative gradient starting at +u, crossing zero at the top (maximum height) and reaching -u on return.
> **(c)** The **area between the line and the time axis** gives the displacement (positive area up, negative area down; net area zero as it returns).

Q26. Two cars are 200 m apart on a straight road and move directly towards each other. Car X travels at a constant 12 m s^{-1}; car Y travels at a constant 8.0 m s^{-1}. **[5]**

1. State the speed of X relative to Y.
2. Find the time before they meet and the distance each has travelled.

> **Answer.** **(a)** Closing (relative) speed = 12 + 8.0 = **20 m s^{-1}**.
> **(b)** Time = 200/20 = **10 s**. X travels 12×10 = **120 m**; Y travels 8.0×10 = **80 m** (sum = 200 m □).

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3 Dynamics

Q27. A block of mass 4.0 kg is released on a slope inclined at 25° to the horizontal. A constant frictional force of 6.0 N acts on the block as it slides down. **[6]**

1. Calculate the resultant force on the block along the slope.
2. Calculate its acceleration.

Answer. **(a)** Weight component down slope = mg sin25° = 4.0×9.81×0.423 = 16.6 N. Resultant = 16.6 - 6.0 = **10.6 N** down the slope.
(b) a = F/m = 10.6/4.0 = **2.7 m s^{-2}**.

Q28. A gun fires bullets of mass 20 g horizontally at a speed of 400 m s^{-1}, at a rate of 5.0 bullets per second. **[5]**

1. Calculate the momentum of one bullet.
2. Calculate the mean force needed to hold the gun steady.

Answer. **(a)** p = mv = 0.020 × 400 = **8.0 kg m s^{-1}**.
(b) Mean force = rate of change of momentum = 8.0 × 5.0 = **40 N** (directed backwards on the gun; the holder supplies 40 N forwards).

Q29. A puck A of mass 0.50 kg moving at 4.0 m s^{-1} strikes a stationary puck B of mass 0.30 kg on a frictionless surface. After the collision A moves at 2.0 m s^{-1} at 30° to its original direction. **[9]**

1. Using conservation of momentum in two perpendicular directions, find the speed of B after the collision.
2. Find the direction in which B moves.

Answer. Take x along A's original direction.

x: $0.50(4.0) = 0.50(2.0)\cos 30° + 0.30v_{Bx} \Rightarrow 2.0 = 0.866 + 0.30v_{Bx} \Rightarrow v_{Bx} = 3.78$ m s^{-1}.

y: $0 = 0.50(2.0)\sin 30° + 0.30v_{By} \Rightarrow 0.30v_{By} = -0.50 \Rightarrow v_{By} = -1.67$ m s^{-1}.

(a) $v_B = \sqrt{3.78^2 + 1.67^2} = \sqrt{17.1}$ = **4.1 m s^{-1}**.
(b) Angle = tan^{-1}(1.67/3.78) = **24° on the opposite side** of the original direction to A.

Q30. Two identical trolleys each of mass m undergo a head-on elastic collision on a frictionless track. Before the collision one moves at speed u and the other is stationary. **[6]**

1. Using conservation of momentum and conservation of kinetic energy, show that the trolleys exchange velocities.
2. State the relationship between relative speed of approach and relative speed of separation for any elastic collision.

> **Answer. (a)** Momentum: $mu = mv_1 + mv_2 \Rightarrow u = v_1 + v_2$. KE: $½mu^2 = ½mv_1^2 + ½mv_2^2 \Rightarrow u^2 = v_1^2 + v_2^2$. From the first, $u^2 = (v_1 + v_2)^2 = v_1^2 + 2v_1v_2 + v_2^2$, so $2v_1v_2 = 0$. Hence one final velocity is 0 and the other is u: they **exchange velocities** ($v_1 = 0$, $v_2 = u$).
> **(b)** Relative speed of approach = relative speed of separation.

Q31. A trolley of mass 2.0 kg moving at 3.0 m s^{-1} collides with and sticks to a stationary trolley of mass 1.0 kg. **[7]**

1. Calculate the common velocity after the collision.
2. Calculate the kinetic energy lost in the collision and state what happens to this energy.

> **Answer. (a)** Momentum: $2.0×3.0 = (2.0 + 1.0)v \Rightarrow v = 6.0/3.0 =$ **2.0 m s^{-1}**.
> **(b)** KE before = $½(2.0)(3.0)^2 = 9.0$ J; KE after = $½(3.0)(2.0)^2 = 6.0$ J. Loss = **3.0 J**, transferred to heat and sound (deformation of the trolleys); momentum is still conserved.

Q32. A resultant force acts on a stationary object of mass 5.0 kg for 4.0 s. The force–time graph is a straight line rising from 0 to 20 N over the 4.0 s. **[6]**

1. Explain how the impulse can be found from the graph, and calculate it.
2. Find the final speed of the object.

> **Answer. (a)** Impulse = change of momentum = area under the force–time graph = $½(4.0)(20)$ = **40 N s**.
>
> **(b)** Impulse = mv - mu = mv (starts at rest) $\Rightarrow v = 40/5.0 =$ **8.0 m s^{-1}**.

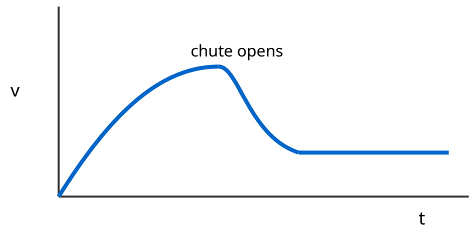
Q33. A cannon of mass 800 kg fires a shell of mass 5.0 kg horizontally at 200 m s^{-1}. **[6]**

1. Calculate the recoil speed of the cannon.
2. Compare the kinetic energy of the shell with that of the cannon, and comment.

> **Answer. (a)** Momentum conserved (initially zero): $800v = 5.0×200 \Rightarrow v = 1000/800 =$ **1.25 m s^{-1}**.
> **(b)** $KE_{shell} = ½(5.0)(200)^2 = 1.0×10^5$ J; $KE_{cannon} = ½(800)(1.25)^2 = 625$ J. The shell has far more KE (ratio 160:1). Momentum is shared equally in magnitude, but $KE = p^2/2m$, so the lighter shell carries most of the energy.

Q34. A skydiver of total mass 85 kg jumps from a stationary balloon. **[7]**

1. Sketch and describe the velocity–time graph from jumping until landing, referring to the forces at each stage (before and after opening the parachute).
2. State the condition for terminal velocity.



Answer. **(a)** Just after jumping, weight > drag, so the diver accelerates but the acceleration decreases as drag rises with speed; the velocity curve rises and levels off at a first (high) terminal velocity. When the parachute opens, drag suddenly exceeds weight, so the diver decelerates (curve falls), then levels off at a lower terminal velocity until landing.
(b) Terminal velocity is reached when the resistive (drag) force equals the weight, so the resultant force is zero and the velocity is constant.

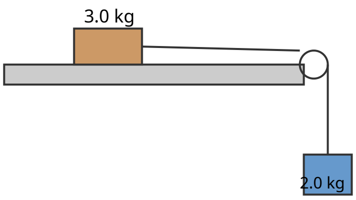
Q35. A person of mass 60 kg stands on bathroom scales inside a lift. **[7]**

1. Find the reading (in newtons) when the lift accelerates upward at 1.5 m s^{-2}.
2. Find the reading when the lift accelerates downward at 1.5 m s^{-2}.
3. State the reading if the lift cable breaks and it falls freely.

Answer. Reading = normal force N. Newton II (up positive): N - mg = ma.
(a) N = m(g + a) = 60(9.81 + 1.5) = **679 N**.
(b) N = m(g - a) = 60(9.81 - 1.5) = **499 N**.
(c) In free fall a = g, so N = m(g - g) = **0 N** (apparent weightlessness).

Q36. A block of mass 3.0 kg rests on a horizontal table and is connected by a light string over a frictionless pulley at the edge to a hanging mass of 2.0 kg. A constant frictional force of 5.0 N acts on the block. **[9]**

1. Calculate the acceleration of the system.
2. Calculate the tension in the string.



> **Answer.** **(a)** Hanging mass: 2.0g - T = 2.0a. Block: T - 5.0 = 3.0a. Adding: 2.0g - 5.0 = 5.0a $\Rightarrow$ a = (19.62 - 5.0)/5.0 = **2.9 m s^{-2}**.
> **(b)** T = 3.0a + 5.0 = 3.0(2.92) + 5.0 = **13.8 N** (check: 2.0(9.81 - 2.92) = 13.8 N □).

Q37. A ball A of mass 2.0 kg moving at 5.0 m s^{-1} makes a head-on elastic collision with a stationary ball B of mass 3.0 kg. **[8]**

1. Using conservation of momentum and the elastic-collision relation (relative approach speed = relative separation speed), find the velocity of each ball after the collision.
2. Verify that kinetic energy is conserved.

> **Answer.** Let final velocities be v_A, v_B.
>
> Momentum: 2.0(5.0) = 2.0v_A + 3.0v_B $\Rightarrow$ 10 = 2v_A + 3v_B. Elastic: v_B - v_A = 5.0. Solving: v_A = **-1.0 m s^{-1}** (rebounds), v_B = **4.0 m s^{-1}**.
> **(b)** KE before = ½(2.0)(5.0)² = 25 J. KE after = ½(2.0)(1.0)² + ½(3.0)(4.0)² = 1.0 + 24 = 25 J. □ Kinetic energy is conserved.

Q38. A ball of mass 0.15 kg travelling horizontally at 12 m s^{-1} strikes a vertical wall and rebounds horizontally at 8.0 m s^{-1}. The contact time is 0.020 s. **[7]**

1. Calculate the change in momentum of the ball.
2. Calculate the average force exerted by the wall on the ball.

> **Answer.** Take the initial direction as positive.
> **(a)** Δp = m(v - u) = 0.15(-8.0 - 12) = 0.15(-20) = **-3.0 kg m s^{-1}** (magnitude 3.0, directed away from the wall).
> **(b)** F = Δp/Δt = 3.0/0.020 = **150 N**.

Q39. Two trolleys are held together with a compressed spring between them; trolley P has mass 1.5 kg and trolley Q has mass 0.50 kg. When released, Q moves off at 3.0 m s^{-1}. **[7]**

1. Find the speed of trolley P.
2. Find the elastic potential energy that was stored in the spring.

> **Answer. (a)** Momentum conserved (initially zero): $1.5v_P = 0.50 \times 3.0 \Rightarrow v_P = 1.5/1.5 =$ **1.0 m s^{-1}** (opposite direction to Q).
> **(b)** Energy = $\frac{1}{2}(1.5)(1.0)^2 + \frac{1}{2}(0.50)(3.0)^2 = 0.75 + 2.25 =$ **3.0 J**.

Q40. State Newton's third law of motion. **[5]**

1. A book rests on a table. Identify the force that pairs with the gravitational pull of the Earth on the book, and the force that pairs with the table's push up on the book.
2. Explain, using Newton's laws, how the exhaust gases from a rocket propel it forward.

> **Answer.** Newton III: when body A exerts a force on body B, body B exerts an equal and opposite force on A (same type, same line of action).
> **(a)** The Earth's pull on the book pairs with the **book's gravitational pull on the Earth**. The table's push up on the book pairs with the **book's push down on the table**. (These are different pairs — not weight and normal force.)
> **(b)** The rocket exerts a backward force on the exhaust gases, expelling them; by Newton III the gases exert an equal forward force on the rocket, giving it forward momentum (momentum is conserved).

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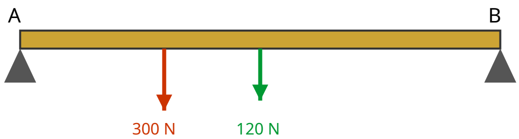
4 Forces, density and pressure

Q41. A uniform beam AB of length 5.0 m and weight 120 N rests horizontally on two supports, one at each end. A load of 300 N is placed 1.5 m from end A. **[8]**

1. By taking moments about A, find the force exerted by the support at B.
2. Hence find the force exerted by the support at A.



Answer. The beam's weight acts at its midpoint, 2.5 m from A.

(a) Moments about A: $R_B(5.0) = 120(2.5) + 300(1.5) = 300 + 450 = 750 \Rightarrow R_B =$ **150 N**.

(b) Vertical equilibrium: $R_A + R_B = 120 + 300 = 420 \Rightarrow R_A = 420 - 150 =$ **270 N**.

Q42. A uniform ladder of weight 200 N and length 8.0 m leans against a smooth vertical wall, making an angle of 60° with the horizontal ground. The ground is rough. **[9]**

1. By taking moments about the foot of the ladder, find the horizontal force exerted by the wall.
2. State the frictional force at the ground and the normal force from the ground.

Answer. The wall is smooth, so it exerts only a horizontal normal force N_W. The weight acts at the centre, 4.0 m along the ladder.

(a) Moments about the foot: $N_W(8.0 \sin 60°) = 200(4.0 \cos 60°) \Rightarrow N_W(6.93) = 200(2.0) = 400 \Rightarrow N_W =$ **57.7 N**.
(b) Horizontal equilibrium: friction = N_W = **57.7 N**. Vertical equilibrium: normal force from ground = weight = **200 N**.

Q43. Explain what is meant by a couple, and define the torque of a couple. A driver turns a steering wheel of diameter 0.36 m by applying two equal, oppositely directed tangential forces of 25 N at opposite ends of a diameter. **[5]**

1. Calculate the torque of the couple.
2. State why the resultant force on the wheel is zero yet it still turns.

Answer. A couple is a pair of equal, opposite, parallel forces whose lines of action do not coincide; its torque = one force × perpendicular distance between them.
(a) Torque = F × d = 25 × 0.36 = **9.0 N m**.
(b) The two forces are equal and opposite, so they add to zero resultant force (no linear acceleration), but because they act along different lines they produce a net turning effect (rotation only).

Q44. An alloy is made by mixing 0.30 kg of metal A (density 2700 kg m^{-3}) with 0.50 kg of metal B (density 8900 kg m^{-3}). Assume the volumes add. **[6]**

1. Calculate the total volume of the alloy.
2. Calculate the density of the alloy.

> **Answer.** **(a)** $V_A = 0.30/2700 = 1.11\times10^{-4}$ m³; $V_B = 0.50/8900 = 5.62\times10^{-5}$ m³. $V = 1.11\times10^{-4} + 5.62\times10^{-5}$ = **1.67×10^{-4} m³**.
> **(b)** $\rho = (0.30 + 0.50)/1.67\times10^{-4} = 0.80/1.67\times10^{-4}$ = **4.8×10^3 kg m^{-3}**.

Q45. A diver is 12 m below the surface of a lake. Atmospheric pressure is 1.01×10^5 Pa and the density of water is 1000 kg m^{-3}. **[6]**

1. Calculate the pressure due to the water alone at this depth.
2. Calculate the total pressure acting on the diver.

> **Answer.** **(a)** $\Delta p = \rho g h = 1000 \times 9.81 \times 12$ = **1.18×10^5 Pa**.
> **(b)** Total = atmospheric + water = $1.01\times10^5 + 1.18\times10^5$ = **2.19×10^5 Pa**.

Q46. A U-tube contains water. Oil, which does not mix with water, is poured into the left arm. When equilibrium is reached, the oil column is 15 cm tall and its lower surface is level with the water in the right arm; the water rises 12 cm above this level on the right. **[6]**

1. State why the pressures at the level of the oil–water boundary are equal in the two arms.
2. Calculate the density of the oil.

> **Answer.** **(a)** Points at the same height in a connected static fluid are at the same pressure; at the oil–water boundary level the pressure from the oil column (left) equals the pressure from the water column (right).
>
> **(b)** $\rho_{oil}g(0.15) = \rho_{water}g(0.12) \Rightarrow \rho_{oil} = 1000 \times 0.12/0.15$ = **800 kg m^{-3}**.

Q47. An object weighs 5.0 N when hung from a spring balance in air, and 3.2 N when fully immersed in water (density 1000 kg m^{-3}). **[8]**

1. Calculate the upthrust and hence the volume of the object.
2. Calculate the density of the object.

> **Answer.** **(a)** Upthrust = 5.0 - 3.2 = 1.8 N. Upthrust = $\rho_w g V \Rightarrow V = 1.8/(1000\times9.81)$ = **1.83×10^{-4} m³**.
> **(b)** Mass = W/g = 5.0/9.81 = 0.510 kg. $\rho = m/V = 0.510/1.83\times10^{-4}$ = **2.8×10^3 kg m^{-3}**.

Q48. A rectangular wooden block of base area 0.040 m^2 and height 0.20 m floats upright in water (density 1000 kg m^{-3}) with 0.15 m of its height submerged. **[7]**

1. Calculate the weight of the block.
2. A load is placed on top so the block just becomes fully submerged. Calculate the weight of the load.

Answer. **(a)** Floating: weight = upthrust = $\rho g V_{sub}$ = 1000×9.81×(0.040×0.15) = 1000×9.81×6.0×10^{-3} = **58.9 N**.
(b) Fully submerged upthrust = ρg(0.040×0.20) = 1000×9.81×8.0×10^{-3} = 78.5 N. Extra upthrust supports the load: W_{load} = 78.5 - 58.9 = **19.6 N**.

Q49. A uniform horizontal rod AB of weight 30 N and length 1.2 m is hinged at A. It is held horizontal by a light string attached at B, which makes an angle of 35° with the rod. A 50 N weight hangs from B. **[8]**

1. By taking moments about A, find the tension in the string.
2. Find the vertical component of the force exerted by the hinge on the rod.

Answer. **(a)** The string's upward (perpendicular) component at B is T sin35°. Moments about A: T sin35°(1.2) = 30(0.6) + 50(1.2) ⇒ T sin35° = (18 + 60)/1.2 = 65 ⇒ T = 65/0.574 = **113 N**.

(b) Vertical equilibrium: V + T sin35° = 30 + 50 ⇒ V = 80 - 65 = **15 N (upward)**.

Q50. A meter rule is balanced horizontally on a knife edge. When a 0.50 N weight is hung at the 10 cm mark, the rule has to be moved so that it balances on the knife edge placed at the 42 cm mark. **[7]**

1. State where the weight of a uniform meter rule acts.
2. By taking moments about the knife edge, find the weight of the rule.

Answer. **(a)** At its **centre of gravity**, the 50 cm mark (uniform rule).
(b) Moments about the 42 cm knife edge: the 0.50 N weight (at 10 cm, i.e. 32 cm away) turns one way; the rule's weight W (at 50 cm, i.e. 8 cm away) turns the other. 0.50 × 0.32 = W × 0.08 ⇒ W = 0.16/0.08 = **2.0 N**.

Q51. A hydraulic jack has a small piston of area 2.0 cm^2 and a large piston of area 50 cm^2. A force of 40 N is applied to the small piston. **[6]**

1. State the principle that makes this device work.
2. Calculate the force produced at the large piston.
3. Explain why this does not violate conservation of energy.

> **Answer.** **(a)** Pressure applied to an enclosed fluid is transmitted equally throughout it, so the pressure is the same on both pistons.
>
> **(b)** $p = F_1/A_1 = F_2/A_2 \Rightarrow F_2 = 40 \times 50/2.0 =$ **1000 N**.
> **(c)** The large piston moves a much smaller distance (25× less) than the small piston, so work in = work out (ignoring friction); force is multiplied but distance is reduced.

Q52. A load of 500 N hangs in equilibrium from a point O, held by a horizontal tie rope and a supporting cable that makes an angle of 40° with the vertical. **[7]**

1. Draw a triangle of forces for point O.
2. Calculate the tension in the cable and the tension in the horizontal tie.

> **Answer.** Three forces act at O: the 500 N weight (down), the cable tension T (up along the cable, 40° from vertical) and the horizontal tie tension H.
> **(a)** A closed triangle: 500 N vertical, H horizontal, T as the hypotenuse.
>
> **(b)** Vertical: $T \cos 40° = 500 \Rightarrow T = 500/0.766 =$ **653 N**. Horizontal: $H = T \sin 40° = 653 \times 0.643 =$ **420 N**.

Q53. Explain, in terms of pressure, why an object experiences an upthrust when immersed in a fluid, and state Archimedes' principle. A solid cube of side 0.10 m is fully submerged in water so that its top face is 0.20 m below the surface. **[7]**

1. Calculate the difference in water pressure between the bottom and top faces.
2. Hence calculate the upthrust, and confirm it equals $\rho g V$.

> **Answer.** Upthrust arises because pressure increases with depth, so the fluid pushes up on the lower face more strongly than it pushes down on the upper face. Archimedes: the upthrust equals the weight of fluid displaced.
> **(a)** Bottom face is 0.30 m deep, top face 0.20 m deep. $\Delta p = \rho g \Delta h = 1000 \times 9.81 \times 0.10 =$ **981 Pa**.
> **(b)** Upthrust $= \Delta p \times \text{area} = 981 \times (0.10)^2 = 981 \times 0.010 =$ **9.81 N**. Also $\rho g V = 1000 \times 9.81 \times (0.10)^3 = 9.81$ N □.

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5 Work, energy and power

Q54. A cyclist and bicycle of total mass 90 kg free-wheel from rest down a hill, descending a vertical height of 25 m. At the bottom the speed is 18 m s^{-1}. **[7]**

1. Calculate the loss in gravitational potential energy.
2. Calculate the kinetic energy at the bottom, and hence the energy dissipated by resistive forces.

Answer. **(a)** $\Delta E_P = mg\Delta h = 90 \times 9.81 \times 25 =$ $\mathbf{2.21 \times 10^4}$ **J**.
(b) $E_K = ½mv^2 = ½(90)(18)^2 = 1.46\times10^4$ J. Energy dissipated = $2.21\times10^4 - 1.46\times10^4 =$ $\mathbf{7.5 \times 10^3}$ **J**.

Q55. A pump raises water from a well 8.0 m deep and delivers it at the surface through a nozzle at a speed of 6.0 m s^{-1}. Water is raised at a rate of 3.0 kg s^{-1}. **[8]**

1. Calculate the power needed to lift the water.
2. Calculate the additional power needed to give the water its kinetic energy.
3. State the minimum input power of the pump.

Answer. **(a)** Rate of gain of PE = $(m/t)gh = 3.0 \times 9.81 \times 8.0 =$ **235 W**.
(b) Rate of gain of KE = $½(m/t)v^2 = ½(3.0)(6.0)^2 =$ **54 W**.

(c) Minimum input power = 235 + 54 = **289 W** ($\approx$ 0.29 kW).

Q56. A car of mass 1200 kg accelerates from 10 m s^{-1} to 30 m s^{-1} along a level road in a distance of 250 m. A constant resistive force of 500 N acts throughout. **[8]**

1. Calculate the gain in kinetic energy.
2. Calculate the work done against resistance.
3. Hence find the average driving force provided by the engine.

Answer. **(a)** $\Delta E_K = ½(1200)(30^2 - 10^2) = ½(1200)(800) =$ $\mathbf{4.8 \times 10^5}$ **J**.
(b) $W_{res} = 500 \times 250 =$ $\mathbf{1.25 \times 10^5}$ **J**.
(c) Work by engine = $\Delta E_K + W_{res} = 4.8\times10^5 + 1.25\times10^5 = 6.05\times10^5$ J. Driving force = $6.05\times10^5/250 =$ $\mathbf{2.4 \times 10^3}$ **N**.

Q57. A ski-lift carries skiers up a slope inclined at 15° to the horizontal at a steady speed of 2.5 m s^{-1}. The total mass being carried is 1800 kg and friction is negligible. **[7]**

1. Calculate the tension in the tow cable.
2. Calculate the power delivered by the motor.

Answer. **(a)** At steady speed, cable tension = weight component along slope = mg sin15° = $1800 \times 9.81 \times 0.259 =$ $\mathbf{4.57 \times 10^3}$ **N**.

(b) $P = Fv = 4.57\times10^3 \times 2.5 =$ $\mathbf{1.14 \times 10^4}$ **W** ($\approx$ 11 kW).

Q58. Derive the equation $P = Fv$ for the power delivered by a constant force F to an object moving at constant velocity v in the direction of the force. [6]

1. Show the derivation from the definitions of work and power.
2. A locomotive develops a useful power of 750 kW while pulling a train at a constant 40 m s^{-1}. Find the total resistive force on the train.

> **Answer.** **(a)** Work done in time t: W = Fs, where s = vt. Power P = W/t = Fs/t = F(vt)/t = **Fv**.
> **(b)** At constant speed the driving force equals the resistance. F = P/v = $750\times10^3/40$ = **1.9×10^4 N**.

Q59. A stone of mass 0.40 kg is projected vertically upward with an initial kinetic energy of 32 J. Air resistance does 6.0 J of work on the stone during its ascent. [6]

1. Using energy conservation, calculate the maximum height reached.
2. State how the maximum height would compare if air resistance were negligible, without recalculating.

> **Answer.** **(a)** Energy available to become PE = 32 - 6.0 = 26 J. mgh = 26 ⇒ h = 26/(0.40 × 9.81) = **6.6 m**.
> **(b)** With no air resistance all 32 J would become PE, giving a **greater** maximum height (about 8.2 m).

Q60. An electric motor rated at 250 W is used to raise a load of mass 30 kg vertically through 4.0 m. The operation takes 8.0 s. [6]

1. Calculate the useful output power.
2. Calculate the efficiency of the motor.

> **Answer.** **(a)** Useful work = mgh = 30 × 9.81 × 4.0 = 1177 J. Useful power = 1177/8.0 = **147 W**.
> **(b)** Efficiency = useful output/total input = 147/250 = 0.59 = **59%**.

Q61. A roller-coaster car of mass 500 kg passes the top of a hill at 8.0 m s^{-1}. It then descends 18 m to the lowest point of the track. A constant frictional force does 2.0×10^4 J of work during the descent. [6]

1. Calculate the speed of the car at the lowest point.

> **Answer.** Energy: $\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mgh - W_f$.
> $\frac{1}{2}(500)v^2 = \frac{1}{2}(500)(8.0)^2 + 500(9.81)(18) - 2.0\times10^4 = 1.60\times10^4 + 8.83\times10^4 - 2.0\times10^4 = 8.43\times10^4$ J. $v^2 = 2(8.43\times10^4)/500 = 337$ ⇒ v = **18.4 m s^{-1}**.

Q62. Derive the formula for kinetic energy $E_K = ½mv^2$ starting from a constant resultant force F accelerating a mass m from rest to speed v. **[6]**

1. Give the derivation using an equation of motion and the definition of work.
2. State the assumption about the force.

> **Answer.** **(a)** Work done by F over distance s is W = Fs = (ma)s. From $v^2 = u^2 + 2as$ with u = 0: as = $v^2/2$. So W = m(as) = m(v^2/2) = **½mv^2**, which is the kinetic energy gained.
> **(b)** The resultant force (and hence acceleration) is assumed constant, and all the work goes into kinetic energy.

Q63. A wind turbine converts the kinetic energy of moving air into electrical energy. Air of density 1.2 kg m^{-3} moves at 9.0 m s^{-1} towards blades that sweep out an area of 1400 m^2. **[7]**

1. Calculate the mass of air passing through the swept area each second.
2. Calculate the maximum power available from this air.

> **Answer.** **(a)** m/t = ρAv = 1.2 × 1400 × 9.0 = **1.51 × 10^4 kg s^{-1}**.
> **(b)** P = ½(m/t)v^2 = ½(1.51×10^4)$(9.0)^2$ = ½(1.51×10^4)(81) = **6.1 × 10^5 W**.

Q64. A 60 kg athlete runs up a flight of stairs of total vertical height 5.0 m in 4.2 s. **[5]**

1. Calculate the useful power developed against gravity.
2. Explain why the athlete's total metabolic power output is considerably greater than this value.

> **Answer.** **(a)** P = mgh/t = 60 × 9.81 × 5.0/4.2 = 2943/4.2 = **701 W**.
> **(b)** Much energy is also used to accelerate limbs, overcome internal friction, and is lost as heat in the muscles; the human body is only partly efficient, so the metabolic (chemical) power input far exceeds the useful mechanical power against gravity.

Q65. A car of mass 1000 kg has a maximum useful engine power of 42 kW. It travels up a slope inclined at 5.0° to the horizontal against a constant resistive force of 600 N. **[6]**

1. Calculate the maximum steady speed the car can maintain up the slope.

> **Answer.** At maximum steady speed the driving force F = total opposing force = mg sin5.0° + 600 = 1000(9.81)(0.0872) + 600 = 855 + 600 = 1455 N.
>
> P = Fv ⇒ v = P/F = 42000/1455 = **28.9 m s^{-1}**.

Q66. A ball of mass 0.20 kg is dropped onto a spring of force constant 800 N m^{-1} from a height of 1.2 m above the top of the spring. Assume no energy is lost. **[7]**

1. Calculate the speed of the ball as it first touches the spring.
2. By equating energy, estimate the maximum compression of the spring (you may neglect the small extra fall during compression).

Answer. **(a)** $v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 1.2}$ = **4.85 m s^{-1}**.
(b) Neglecting the extra fall, the ball's energy on reaching the spring = mgh = 0.20×9.81×1.2 = 2.35 J stored as $\frac{1}{2}kx^2$: $x = \sqrt{2E/k} = \sqrt{2 \times 2.35/800} = \sqrt{5.88 \times 10^{-3}}$ = **7.7×10^{-2} m** (about 7.7 cm).

6 Deformation of solids

Q67. A steel wire of natural length 2.5 m and diameter 0.80 mm is stretched by a force of 65 N. The Young modulus of steel is 2.0×10^{11} Pa. **[7]**

1. Calculate the stress in the wire.
2. Calculate the extension produced.

Answer. $A = \pi(0.40\times10^{-3})^2 = 5.03\times10^{-7}$ m².
(a) Stress = F/A = $65/5.03\times10^{-7}$ = **1.29×10^8 Pa**.
(b) Strain = stress/E = $1.29\times10^8/2.0\times10^{11} = 6.46\times10^{-4}$. Extension = strain × L = $6.46\times10^{-4} \times 2.5$ = **1.6×10^{-3} m** (1.6 mm).

Q68. Describe an experiment to determine the Young modulus of a metal in the form of a long wire. **[7]**

1. State the measurements taken and the instrument used for each.
2. Explain how a graph is used to obtain the Young modulus, and why a long thin wire is used.

Answer. **(a)** Measure the original length L (metre rule / tape), the diameter d at several points (micrometer) to find $A = \pi d^2/4$, the added load F (known masses × g), and the extension e for each load (marker + travelling microscope or a Vernier scale).
(b) Plot F against e: within the limit of proportionality this is a straight line of gradient F/e. Since E = (F/e)(L/A), E = gradient × L/A. A long thin wire gives a larger, more easily measured extension (e = FL/AE), reducing the percentage uncertainty.

Q69. A copper wire and a steel wire have the same length and the same diameter and are joined end to end so they hang vertically in series, supporting a load of 40 N. $E_{copper} = 1.2 \times 10^{11}$ Pa, $E_{steel} = 2.0 \times 10^{11}$ Pa. **[7]**

1. Explain why the two wires experience the same stress but different strains.
2. Find the ratio of the extension of the copper wire to that of the steel wire.

Answer. **(a)** In series each wire carries the same tension (40 N) and has the same area, so stress (= F/A) is equal. Since strain = stress/E and E differs, the strains differ (copper strains more because its E is smaller).

(b) Same length and stress: e = (stress/E)L, so $e \in 1/E$. $e_{Cu}/e_{steel} = E_{steel}/E_{Cu} = 2.0/1.2$ = **1.7**.

Q70. A spring obeys Hooke's law and extends by 4.0 cm when a force of 5.0 N is applied. **[7]**

1. Calculate the spring constant.
2. Calculate the elastic potential energy stored at this extension.
3. Calculate the extra energy needed to stretch it a further 2.0 cm.

> **Answer. (a)** $k = F/x = 5.0/0.040 =$ **$125\ N\ m^{-1}$**.
> **(b)** $E = ½kx^2 = ½(125)(0.040)^2 =$ **0.10 J**.
> **(c)** At 6.0 cm: $E' = ½(125)(0.060)^2 = 0.225$ J. Extra = 0.225 - 0.10 = **0.13 J** (more than the first because energy $\in x^2$).

Q71. Two identical springs, each of spring constant $50\ N\ m^{-1}$, support a load of 12 N. **[6]**

1. Calculate the extension if the springs are connected in parallel (side by side).
2. Calculate the extension if the springs are connected in series (end to end).

> **Answer. (a)** Parallel: $k_{eff} = 50 + 50 = 100\ N\ m^{-1}$. $x = F/k = 12/100 =$ **0.12 m**.
> **(b)** Series: $1/k_{eff} = 1/50 + 1/50 \Rightarrow k_{eff} = 25\ N\ m^{-1}$. $x = 12/25 =$ **0.48 m**.

Q72. The force–extension graph for a metal wire is a straight line from the origin up to a force of 30 N at an extension of 1.5 mm (the limit of proportionality), after which it curves and the wire eventually breaks at an extension of 6.0 mm. **[7]**

1. Calculate the spring constant (stiffness) of the wire in its linear region.
2. Calculate the elastic potential energy stored at the limit of proportionality.
3. State whether the deformation at 6.0 mm is elastic or plastic, with a reason.

> **Answer. (a)** $k = F/x = 30/1.5\times10^{-3} =$ **$2.0 \times 10^4\ N\ m^{-1}$**.
> **(b)** $E = ½Fx = ½(30)(1.5\times10^{-3}) =$ **2.25×10^{-2} J**.
> **(c) Plastic**: 6.0 mm is well beyond the elastic limit, so the wire would not return to its original length if unloaded (permanent deformation).

Q73. Define stress, strain and the Young modulus, and state the unit of each. **[6]**

1. Give the three definitions and units.
2. Explain why strain has no unit.

> **Answer. (a)** Stress = force per unit cross-sectional area (F/A), unit Pa ($N\ m^{-2}$). Strain = extension per unit original length (e/L), no unit. Young modulus E = stress/strain, unit Pa.
> **(b)** Strain is a ratio of two lengths (extension ÷ original length), so the units cancel and it is a pure number.

Q74. A nylon rope of cross-sectional area 1.2×10^{-4} m² and Young modulus 3.0×10^{9} Pa hangs vertically with length 20 m. A climber of weight 700 N hangs from its lower end. **[6]**

1. Calculate the stress in the rope.
2. Calculate the extension of the rope.

> **Answer.** **(a)** Stress = F/A = $700/1.2\times10^{-4}$ = **5.8×10^{6} Pa**.
> **(b)** Strain = stress/E = $5.83\times10^{6}/3.0\times10^{9}$ = 1.94×10^{-3}. Extension = strain × L = 1.94×10^{-3} × 20 = **3.9×10^{-2} m** (3.9 cm).

Q75. Distinguish between elastic and plastic deformation, and explain the meaning of the elastic limit. **[6]**

1. Give the distinction and define the elastic limit.
2. State what is meant by the limit of proportionality and how it relates to Hooke's law.

> **Answer.** **(a)** Elastic deformation: the material returns to its original shape/size when the load is removed. Plastic deformation: the material keeps a permanent change of shape after unloading. The **elastic limit** is the maximum load (or stress) beyond which the deformation becomes permanent (plastic).
> **(b)** The limit of proportionality is the point up to which extension is proportional to load (the force–extension graph is a straight line); Hooke's law (F = kx) holds only up to this point.

Q76. A stress–strain graph is plotted for a ductile metal. It rises linearly, then reaches a peak (the ultimate tensile stress) before the metal necks and fractures. **[5]**

1. Explain what happens to the specimen physically after the peak of the graph.
2. State what the area under a stress–strain graph, up to fracture, represents.

> **Answer.** **(a)** Beyond the peak the metal undergoes plastic flow: a narrow region 'necks' (reduces in cross-section), so the stress needed appears to fall until the specimen fractures there.
> **(b)** The area under the stress–strain graph represents the **energy per unit volume** (work done per unit volume) needed to stretch the material — a measure of its toughness.

Q77. A load is applied to a wire and then removed. The loading and unloading lines on a force–extension graph do not coincide, and the unloading line returns to a non-zero extension. **[5]**

1. Explain what this tells you about how the wire was loaded.
2. State what the area between the loading and unloading lines represents.

> **Answer.** **(a)** The wire was loaded **beyond its elastic limit**, so it deformed plastically and did not return to its original length — hence the permanent (non-zero) extension on unloading.
> **(b)** The area enclosed between the loading and unloading curves represents the energy dissipated (as heat) in permanently deforming the wire.

Q78. A vertical steel wire supports a mass. When the mass is 2.0 kg the extension is 0.60 mm; the wire obeys Hooke's law over this range. [6]

1. Calculate the spring constant of the wire.
2. Predict the extension for a 3.5 kg mass, and state the assumption made.

Answer. **(a)** Force = 2.0 × 9.81 = 19.6 N. k = F/x = $19.6/0.60\times10^{-3}$ = $\mathbf{3.27 \times 10^{4}}$ **N** $\mathbf{m^{-1}}$.
(b) Force = 3.5 × 9.81 = 34.3 N. x = F/k = $34.3/3.27\times10^{4}$ = $\mathbf{1.05 \times 10^{-3}}$ **m** (1.05 mm). Assumption: the wire still obeys Hooke's law (load within the limit of proportionality).

7 Waves

Q79. A progressive sound wave of frequency 256 Hz travels through air at 340 m s^{-1}. **[7]**

1. Calculate the wavelength.
2. Calculate the phase difference, in radians, between two points 0.44 m apart along the direction of travel.

Answer. **(a)** $\lambda = v/f = 340/256 =$ **1.33 m**.
(b) Phase difference = $(\Delta x/\lambda) \times 2\pi = (0.44/1.328) \times 2\pi = 0.331 \times 2\pi =$ **2.1 rad** (about 0.66π).

Q80. A cathode-ray oscilloscope (CRO) displays the waveform of a sound. The time-base is set to 2.0 ms per division and the trace shows one complete cycle across 4.0 divisions. The y-gain is 0.50 V per division and the trace has a peak height of 3.0 divisions from the centre line. **[7]**

1. Calculate the frequency of the sound.
2. State the peak voltage of the signal.

Answer. **(a)** Period $T = 4.0 \times 2.0\times10^{-3} = 8.0\times10^{-3}$ s. $f = 1/T = 1/8.0\times10^{-3} =$ **125 Hz**.
(b) Peak voltage = $3.0 \times 0.50 =$ **1.5 V**.

Q81. A point source emits sound uniformly in all directions with a power of 1.6 W. **[7]**

1. Calculate the intensity of the sound at a distance of 4.0 m from the source.
2. State how the amplitude of the wave at 8.0 m compares with that at 4.0 m, justifying your answer.

Answer. **(a)** $I = P/(4\pi r^2) = 1.6/(4\pi \times 4.0^2) = 1.6/201 =$ **8.0×10^{-3} W m^{-2}**.

(b) Doubling r to 8.0 m reduces I by a factor 4 (inverse-square). Since $I \in \text{amplitude}^2$, the amplitude falls by a factor $\sqrt{4} = 2$, i.e. to **half** its value at 4.0 m.

Q82. Compare transverse and longitudinal waves. **[6]**

1. State the difference in terms of the direction of oscillation relative to the direction of energy transfer, giving one example of each.
2. State which type can be polarised, and why.

Answer. **(a)** In a transverse wave the oscillations are perpendicular to the direction of energy travel (e.g. electromagnetic waves, waves on a string); in a longitudinal wave the oscillations are parallel to the direction of travel, producing compressions and rarefactions (e.g. sound).
(b) Only **transverse** waves can be polarised, because polarisation restricts oscillations to a single plane — this is only possible when the oscillation is perpendicular to the direction of travel (longitudinal oscillations are already along one axis).

Q83. A source of sound emits a note of frequency 480 Hz. It moves directly towards a stationary observer at 30 m s^{-1}. The speed of sound in air is 340 m s^{-1}. **[7]**

1. Calculate the frequency heard by the observer.
2. State and explain qualitatively what the observer hears just after the source passes.

Answer. **(a)** Approaching: $f_o = f_s v/(v - v_s)$ = 480 × 340/(340 - 30) = 480 × 340/310 = **527 Hz**.
(b) After passing, the source recedes, so $f_o = f_s v/(v + v_s)$ = 480 × 340/370 = 441 Hz — the pitch drops suddenly to a lower, steady value.

Q84. Light is plane-polarised by passing through a polarising filter. It then passes through a second filter (analyser) whose transmission axis is at 30° to that of the first. **[6]**

1. State Malus's law and use it to find the fraction of the intensity transmitted by the second filter.
2. The analyser is now rotated to 90°. State the transmitted intensity and explain.

Answer. **(a)** Malus: $I = I_0 \cos^2\theta$, where θ is the angle between the polarisation direction and the analyser axis. Fraction = $\cos^2 30°$ = $(0.866)^2$ = **0.75** (75%).
(b) At 90°, $\cos^2 90°$ = 0, so **no light** is transmitted — the filters are 'crossed' and the transmitted electric-field component is zero.

Q85. State the approximate range of wavelengths in free space for the principal regions of the electromagnetic spectrum. **[7]**

1. List, in order of increasing wavelength, the regions from γ-rays to radio waves with an approximate wavelength for each.
2. State what all electromagnetic waves have in common in free space, and the range visible to the eye.

Answer. **(a)** γ-rays (< 10^{-11} m); X-rays (10^{-11}–10^{-9} m); ultraviolet (10^{-9}–4×10^{-7} m); visible (4–7 × 10^{-7} m); infrared (7×10^{-7}–10^{-3} m); microwaves (10^{-3}–10^{-1} m); radio waves (> 10^{-1} m).
(b) All are transverse and travel at the same speed $c = 3.0\times10^8$ m s^{-1} in free space; the eye sees **400–700 nm**.

Q86. A wave travelling along a string is described by a displacement–time graph for one particle showing simple harmonic oscillation of amplitude 5.0 mm and period 0.40 s. **[7]**

1. State the frequency of the wave.
2. If the wave speed is 12 m s^{-1}, calculate the wavelength.
3. Explain the difference between the displacement–time graph for one particle and a displacement–position graph of the whole wave at one instant.

> **Answer.** **(a)** f = 1/T = 1/0.40 = **2.5 Hz**.
> **(b)** λ = v/f = 12/2.5 = **4.8 m**.
> **(c)** A displacement–time graph shows how one particle's displacement varies with time (period T on the axis); a displacement–position graph is a snapshot of all particles at one instant (wavelength λ on the axis). They can look similar but the axes represent different quantities.

Q87. Unpolarised light of intensity I_0 passes through two polarising filters whose axes are parallel. **[7]**

1. State the intensity transmitted after the first filter (in terms of I_0).
2. A third filter is inserted between them with its axis at 45° to both. Find the final intensity in terms of I_0.

> **Answer.** **(a)** An ideal polariser transmits half of unpolarised light: $\mathbf{I_0/2}$ (now plane-polarised along the first axis).
> **(b)** After the middle filter (at 45°): $(I_0/2)\cos^2 45° = (I_0/2)(0.5) = I_0/4$. After the last filter (at 45° to the middle one): $(I_0/4)\cos^2 45°$ = $\mathbf{I_0/8}$. (Inserting a filter lets light through a previously crossed pair.)

Q88. Explain what is meant by the statement that a progressive wave transfers energy. **[5]**

1. Describe, for a wave on the surface of water, what is transferred and what is not.
2. State how the intensity of a progressive wave depends on its amplitude.

> **Answer.** **(a)** The wave transfers energy through the medium in the direction of travel, but the water particles themselves only oscillate about fixed mean positions — there is no net transfer of matter (a floating object bobs up and down but does not travel with the wave).
>
> **(b)** Intensity is proportional to the square of the amplitude ($I \in A^2$).

Q89. A sound source and a stationary observer are used to demonstrate the Doppler effect. The observed frequency is 1.10 times the source frequency. **[6]**

1. State whether the source is approaching or receding, and explain.
2. Calculate the speed of the source (speed of sound = 340 m s^{-1}).

> **Answer.** **(a)** Observed frequency is higher than the source frequency, so the source is **approaching** (the wavefronts are compressed in front of a moving source).
>
> **(b)** $f_o = f_s v/(v - v_s) \Rightarrow 1.10 = 340/(340 - v_s) \Rightarrow 340 - v_s = 340/1.10 = 309 \Rightarrow v_s =$ **31 m s^{-1}**.

Q90. State what is meant by the polarisation of a wave, and describe how a polarising filter affects unpolarised light passing through it. **[6]**

1. Define polarisation.
2. Describe an everyday application of polarising filters and the physics behind it.

Answer. **(a)** Polarisation is the restriction of the oscillations of a transverse wave to a single plane containing the direction of travel. A polarising filter transmits only the component of the electric field parallel to its transmission axis, so emerging light is plane-polarised (and about half the intensity for unpolarised input).
(b) Polarising sunglasses reduce glare: light reflected from horizontal surfaces (water, roads) is partially horizontally polarised, and the glasses' vertical axis blocks this component, cutting the reflected glare.

8 Superposition

Q91. In a double-slit experiment, coherent light of wavelength 590 nm passes through two slits separated by 0.45 mm. Fringes are observed on a screen 2.2 m from the slits. **[7]**

1. Calculate the separation of adjacent bright fringes.
2. State and explain what happens to the fringe separation if the slit separation is increased.

Answer. **(a)** $x = \lambda D/a = (590\times10^{-9} \times 2.2)/0.45\times10^{-3} = 1.298\times10^{-6}/4.5\times10^{-4} = \mathbf{2.9 \times 10^{-3}}$ **m** (2.9 mm).
(b) Since $x = \lambda D/a$, increasing a **decreases** x — the fringes move closer together.

Q92. Explain what is meant by the terms coherence and interference, and state the conditions needed to observe a stable two-source interference pattern with light. **[6]**

1. Define coherence and interference.
2. List the conditions for observable, stable fringes.

Answer. **(a)** Two sources are coherent if they have a constant phase difference (and the same frequency). Interference is the superposition of waves from two (or more) sources to give a resultant of larger or smaller amplitude (constructive/destructive).
(b) The sources must be coherent (constant phase difference, same frequency), have approximately equal amplitudes, and (for transverse waves such as light) be polarised in the same plane; the path difference should be comparable to the wavelength so fringes are resolvable.

Q93. A diffraction grating has 500 lines per millimetre. It is illuminated normally by monochromatic light of wavelength 6.0×10^{-7} m. **[9]**

1. Calculate the grating spacing.
2. Calculate the angle of diffraction of the second-order maximum.
3. Determine the total number of bright maxima observable.

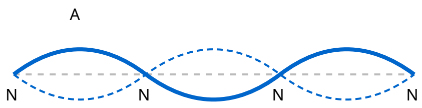
Answer. **(a)** $d = 1/(500\times10^{3}) = \mathbf{2.0 \times 10^{-6}}$ **m**.

(b) $d \sin\theta = n\lambda$: $\sin\theta = 2 \times 6.0\times10^{-7}/2.0\times10^{-6} = 0.60 \Rightarrow \theta =$ **37°**.

(c) Maximum order: $n \leq d/\lambda = 2.0\times10^{-6}/6.0\times10^{-7} = 3.3 \Rightarrow n_{max} = 3$. Orders 0, ±1, ±2, ±3 ⇒ **7 maxima**.

Q94. A stationary wave is set up on a stretched string of length 1.2 m fixed at both ends. The string vibrates in its third harmonic (three loops). **[8]**

1. Sketch the shape of the string, marking nodes (N) and antinodes (A).
2. Calculate the wavelength.
3. If the wave speed on the string is 240 m s^{-1}, calculate the frequency.



> **Answer.** **(a)** Three loops between the fixed ends: N–A–N–A–N–A–N (nodes at both ends and at two interior points, antinodes at the centre of each loop).
>
> **(b)** Three half-wavelengths fit the string: 3(λ/2) = 1.2 ⇒ λ = **0.80 m**.
> **(c)** f = v/λ = 240/0.80 = **300 Hz**.

Q95. A tube closed at one end and open at the other has a length of 0.60 m. The speed of sound in air is 340 m s^{-1}. (Neglect end corrections.) **[8]**

1. Sketch the displacement pattern of the fundamental and state where the node and antinode are.
2. Calculate the fundamental frequency.
3. State the frequency of the next possible harmonic.

> **Answer.** **(a)** A closed–open tube has a displacement **node at the closed end** and an **antinode at the open end**; the fundamental fits a quarter-wavelength.
>
> **(b)** L = λ/4 ⇒ λ = 4L = 2.4 m. f_1 = v/λ = 340/2.4 = **142 Hz**.
> **(c)** A closed tube supports only odd harmonics, so the next is the third harmonic: 3 × 142 = **425 Hz**.

Q96. Microwaves from a source are directed at a metal plate, and a probe detector is moved along the line between the source and plate. The detector signal rises and falls, with minima 15 mm apart. **[7]**

1. Explain why a stationary wave is formed in this region.
2. Calculate the wavelength and hence the frequency of the microwaves.

> **Answer.** **(a)** The incident microwaves reflect from the metal plate; the reflected wave superposes on the incident wave (same frequency, travelling in opposite directions), producing a stationary wave with fixed nodes and antinodes.
> **(b)** Adjacent minima (nodes) are half a wavelength apart, so λ = 2 × 15 = 30 mm = 0.030 m. f = c/λ = $3.0\times10^8/0.030$ = **1.0×10^{10} Hz**.

Q97. Explain, using the principle of superposition, how a stationary wave differs from a progressive wave. **[6]**

1. State the principle of superposition.
2. Give two differences between a stationary wave and a progressive wave (energy and amplitude).

Answer. **(a)** When two or more waves meet at a point, the resultant displacement equals the vector sum of the individual displacements at that point.
(b) (i) A progressive wave transfers energy along its direction of travel; a stationary wave stores energy and transfers none along the wave. (ii) In a progressive wave every point has the same amplitude; in a stationary wave the amplitude varies with position from zero at nodes to a maximum at antinodes. (Also: all points between adjacent nodes oscillate in phase in a stationary wave.)

Q98. Light of wavelength 480 nm illuminates a double slit, giving fringes of separation 1.8 mm on a screen 1.5 m away. **[7]**

1. Calculate the slit separation.
2. The blue light is replaced by red light of wavelength 660 nm. Calculate the new fringe separation.

Answer. **(a)** $a = \lambda D/x = (480\times10^{-9} \times 1.5)/1.8\times10^{-3} = 7.2\times10^{-7}/1.8\times10^{-3} =$ **4.0×10^{-4} m** (0.40 mm).

(b) $x \in \lambda$ (same a, D): $x' = 1.8 \times 660/480 =$ **2.5 mm**.

Q99. Explain the meaning of diffraction, and describe an experiment using a ripple tank to show how the amount of diffraction depends on the gap width compared with the wavelength. **[6]**

1. Define diffraction.
2. Describe the observations for a gap much wider than, and a gap comparable to, the wavelength.

Answer. **(a)** Diffraction is the spreading of a wave as it passes through a gap or around an obstacle.
(b) In a ripple tank, straight water waves pass through a gap between two barriers. When the gap is much wider than the wavelength, the waves pass through nearly straight with only slight spreading at the edges; when the gap is about the same size as the wavelength, the waves spread out in strong semicircular arcs beyond the gap.

Q100. Describe how a diffraction grating can be used to determine the wavelength of monochromatic light. **[6]**

1. State the apparatus and measurements taken.
2. State the equation used and how the wavelength is calculated.

Answer. **(a)** Direct a narrow beam of the monochromatic light normally at a grating of known spacing d (from lines per metre). Measure the angle θ of a diffracted order (e.g. first order) using a rotating table/protractor, ideally measuring the angle to the order on both sides and halving to reduce error.
(b) Use $d \sin\theta = n\lambda$, so $\lambda = d \sin\theta/n$. Repeating for several orders and averaging improves reliability.

Q101. Two coherent loudspeakers face an area and produce sound of the same frequency in phase. **[6]**
A listener walking across the area hears alternating loud and quiet regions.

1. Explain, in terms of path difference, why loud and quiet regions occur.
2. State the path difference (in terms of wavelength λ) at a point of minimum loudness.

Answer. **(a)** Sound from the two speakers travels different distances to a given point. Where the path difference is a whole number of wavelengths the waves arrive in phase and interfere constructively (loud); where it is an odd number of half-wavelengths they arrive out of phase and interfere destructively (quiet).
(b) Minimum loudness (destructive): path difference = $(n + ½)\lambda$, i.e. $½\lambda$, $3½\ldots\lambda$ — an odd number of half-wavelengths.

Q102. A grating spectrum is produced when white light passes through a diffraction grating. **[6]**

1. Explain why the central (zero-order) image is white but the higher orders are spectra.
2. State which colour is diffracted through the largest angle in a given order, with a reason.

Answer. **(a)** At the zero order ($n = 0$) the path difference is zero for all wavelengths, so all colours arrive in phase and overlap to give white. For $n \geq 1$, the diffraction angle depends on λ ($d \sin\theta = n\lambda$), so different colours emerge at different angles and separate into a spectrum.

(b) **Red** is diffracted most in each order, because it has the longest wavelength and $\sin\theta \in \lambda$.

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9 Electricity

Q103. A copper wire carries a current of 3.5 A. The wire has a cross-sectional area of 1.5×10^{-6} m^2 and the number density of free electrons is 8.5×10^{28} m^{-3}. **[7]**

1. Calculate the mean drift velocity of the electrons.
2. Comment on the size of your answer and explain how a lamp can light almost instantly.

> **Answer.** **(a)** $v = I/(nAq) = 3.5/(8.5\times10^{28} \times 1.5\times10^{-6} \times 1.60\times10^{-19}) = 3.5/2.04\times10^{4}$ = $\mathbf{1.7 \times 10^{-4}}$ **m** $\mathbf{s^{-1}}$.
> **(b)** The drift speed is tiny (< 1 mm s^{-1}). A lamp lights almost at once because the electric field is set up along the wire at close to the speed of light, so free electrons everywhere — including in the lamp — begin to drift almost simultaneously.

Q104. A charge of 90 C passes through a lamp in 1.0 minute. The lamp is connected to a 12 V supply. **[7]**

1. Calculate the current in the lamp.
2. Calculate the power dissipated and the energy transferred in this time.

> **Answer.** **(a)** I = Q/t = 90/60 = **1.5 A**.
> **(b)** P = VI = 12 × 1.5 = **18 W**. Energy = Pt = 18 × 60 = $\mathbf{1.08 \times 10^{3}}$ **J** (equivalently W = QV = 90 × 12 = 1080 J).

Q105. Define the resistance of a component, and state Ohm's law. **[6]**

1. Give both statements precisely.
2. A component obeys V = IR with R = 15 Ω. Calculate the power dissipated when the current is 0.40 A, using two different formulae as a check.

> **Answer.** **(a)** Resistance = potential difference across a component divided by the current through it (R = V/I). Ohm's law: the current in a metallic conductor is directly proportional to the potential difference across it, provided physical conditions (especially temperature) are constant.
> **(b)** $P = I^2R = (0.40)^2(15)$ = **2.4 W**. Check: V = IR = 6.0 V, P = VI = 6.0 × 0.40 = 2.4 W ▯.

Q106. A metal wire has resistance 4.0 Ω. A second wire of the same material has twice the length and half the diameter of the first. **[7]**

1. Explain how resistance depends on length and cross-sectional area.
2. Calculate the resistance of the second wire.

> **Answer.** **(a)** R = ρL/A: resistance is proportional to length and inversely proportional to cross-sectional area.
>
> **(b)** Twice the length ⇒ ×2. Half the diameter ⇒ area ×¼ ⇒ R ×4. Combined factor = 2 × 4 = 8. R = 8 × 4.0 = **32 Ω**.

Q107. Sketch and describe the I–V characteristic of a filament lamp. **[6]**

1. Describe the shape of the graph.
2. Explain, in terms of the behaviour of the metal, why the resistance increases as the current increases.

> **Answer. (a)** The graph is an S-shaped curve through the origin: for small currents it is nearly straight, but the gradient decreases (the curve bends towards the V-axis) as V and I increase, showing that V/I (resistance) rises.
> **(b)** A larger current dissipates more power, heating the filament. The higher temperature makes the metal ions vibrate more, so free electrons collide with them more frequently, impeding the flow — hence the resistance increases.

Q108. A nichrome wire of length 1.5 m and diameter 0.30 mm has a resistivity of 1.1×10^{-6} Ω m. **[7]**

1. Calculate the resistance of the wire.
2. Calculate the power dissipated when a p.d. of 6.0 V is applied across it.

> **Answer.** $A = \pi(0.15\times10^{-3})^2 = 7.07\times10^{-8}$ m².
> **(a)** $R = \rho L/A = (1.1\times10^{-6} \times 1.5)/7.07\times10^{-8} = 1.65\times10^{-6}/7.07\times10^{-8} =$ **23 Ω**.
> **(b)** $P = V^2/R = 6.0^2/23.3 = 36/23.3 =$ **1.5 W**.

Q109. Explain how the resistance of (a) a thermistor and (b) a light-dependent resistor (LDR) changes with external conditions, and describe one use of each. **[6]**

1. State the behaviour of each component.
2. State a practical use of each.

> **Answer. (a)** A thermistor (negative temperature coefficient) has resistance that **decreases as temperature increases**, because more charge carriers are freed. An LDR has resistance that **decreases as light intensity increases**, for the same reason (light frees charge carriers).
> **(b)** Thermistor: temperature sensing / thermostat control. LDR: automatic lighting / light-level switching (e.g. street lamps).

Q110. An electric kettle is rated at 2.2 kW, 230 V. **[8]**

1. Calculate the operating current and the resistance of the element.
2. Calculate the time taken to raise the temperature of 0.50 kg of water by 80 K, assuming all the energy heats the water. (Specific heat capacity of water = 4200 J kg^{-1} K^{-1}.)

> **Answer. (a)** $I = P/V = 2200/230 =$ **9.6 A**. $R = V^2/P = 230^2/2200 =$ **24 Ω**.
> **(b)** Energy needed $= mc\Delta\theta = 0.50 \times 4200 \times 80 = 1.68\times10^5$ J. $t = E/P = 1.68\times10^5/2200 =$ **76 s**.

Q111. Two wires, X and Y, are made of the same material and have the same length. X has twice the cross-sectional area of Y. They are connected in parallel across a battery. **[7]**

1. Compare the resistances of X and Y.
2. Compare the currents in, and the drift velocities of electrons in, the two wires.

Answer. **(a)** $R = \rho L/A$. Same ρ and L; X has twice the area, so $R_X = ½R_Y$ (X has half the resistance).
(b) Same p.d. across each (parallel). $I = V/R$, so $I_X = 2I_Y$ (X carries twice the current). Drift velocity $v = I/(nAq)$: $I_X/A_X = (2I_Y)/(2A_Y) = I_Y/A_Y$, so the **drift velocities are equal**.

Q112. Define the potential difference across a component in terms of energy. **[5]**

1. Give the definition and the defining equation.
2. A charge of 25 mC passes through a resistor and transfers 0.30 J of energy to it. Calculate the p.d. across the resistor.

Answer. **(a)** The p.d. across a component is the electrical energy transferred to other forms per unit charge passing through it: $V = W/Q$.
(b) $V = W/Q = 0.30/25\times10^{-3}$ = **12 V**.

Q113. A student investigates how the resistance of a length of resistance wire varies and obtains a current of 0.25 A when the p.d. is 5.0 V. **[6]**

1. Calculate the resistance.
2. The temperature of the wire then rises and the current falls to 0.22 A at the same p.d. Calculate the new resistance and comment on whether the wire is ohmic.

Answer. **(a)** $R = V/I = 5.0/0.25$ = **20 Ω**.
(b) $R' = 5.0/0.22$ = **23 Ω**. The resistance changed when conditions (temperature) changed, so over this range it is **not behaving ohmically** — Ohm's law requires constant temperature.

Q114. A semiconductor diode is connected in series with a resistor and a variable d.c. supply so that it is forward biased. **[6]**

1. Sketch the I–V characteristic of the diode for both forward and reverse bias.
2. Explain the purpose of the series resistor.

Answer. **(a)** Forward bias: negligible current until about 0.6 V, then the current rises very steeply. Reverse bias: only a tiny (effectively zero) current for all reverse voltages up to breakdown.
(b) Once conducting, the diode's resistance is very low, so the current could rise dangerously; the series resistor limits the current to a safe value and takes the extra p.d.

Q115. A 12 V car battery delivers a current of 40 A to a starter motor for 3.0 s. **[6]**

1. Calculate the charge delivered and the energy transferred.
2. The motor is 60% efficient. Calculate the useful mechanical energy output.

Answer. **(a)** Q = It = 40 × 3.0 = **120 C**. Energy = VIt = 12 × 40 × 3.0 = $\mathbf{1.44 \times 10^3}$ **J**.
(b) Useful output = 0.60 × 1.44×10^3 = **864 J**.

Q116. State what is meant by an electric current in terms of charge carriers, and explain what is meant by saying charge is quantised. **[6]**

1. Define electric current and give the quantisation of charge.
2. A current of 0.16 A flows for 5.0 s. Calculate the number of electrons that pass a point.

Answer. **(a)** Current is the rate of flow of charge ($I = \Delta Q/\Delta t$). Charge is quantised: it exists only in whole-number multiples of the elementary charge e = 1.60×10^{-19} C.
(b) Q = It = 0.16 × 5.0 = 0.80 C. Number = Q/e = $0.80/1.60\times10^{-19}$ = $\mathbf{5.0 \times 10^{18}}$ **electrons**.

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10 D.C. circuits

Q117. A battery of e.m.f. 9.0 V and internal resistance 1.5 Ω is connected to an external resistor of 6.0 Ω. **[7]**

1. Calculate the current in the circuit.
2. Calculate the terminal potential difference and the power dissipated inside the battery.

> **Answer.** **(a)** I = ε/(R + r) = 9.0/(6.0 + 1.5) = 9.0/7.5 = **1.2 A**.
> **(b)** Terminal p.d. = IR = 1.2 × 6.0 = **7.2 V** (or ε - Ir = 9.0 - 1.8). Power in internal resistance = I^2r = $(1.2)^2(1.5)$ = **2.2 W**.

Q118. State Kirchhoff's first and second laws, and state the conservation principle underlying each. **[6]**

1. Give both laws and the corresponding conservation law.
2. At a junction, currents of 2.0 A and 1.5 A flow in, and currents of 0.80 A and I flow out. Find I.

> **Answer.** **(a)** First law: the sum of currents into a junction equals the sum of currents out (conservation of charge). Second law: around any closed loop, the sum of the e.m.f.s equals the sum of the p.d.s (conservation of energy).
>
> **(b)** In = out: 2.0 + 1.5 = 0.80 + I ⇒ I = 3.5 - 0.80 = **2.7 A**.

Q119. Three resistors of 2.0 Ω, 3.0 Ω and 6.0 Ω are available. **[8]**

1. Calculate the combined resistance if all three are connected in series.
2. Calculate the combined resistance if all three are connected in parallel.
3. Calculate the combined resistance if the 3.0 Ω and 6.0 Ω are in parallel and this combination is in series with the 2.0 Ω.

> **Answer.** **(a)** Series: 2.0 + 3.0 + 6.0 = **11 Ω**.
>
> **(b)** Parallel: 1/R = 1/2.0 + 1/3.0 + 1/6.0 = 0.5 + 0.333 + 0.167 = 1.0 ⇒ R = **1.0 Ω**.
>
> **(c)** 3.0 Ω ▯ 6.0 Ω: 1/R = 1/3.0 + 1/6.0 = 0.5 ⇒ R = 2.0 Ω. In series with 2.0 Ω: 2.0 + 2.0 = **4.0 Ω**.

Q120. A battery of e.m.f. E and internal resistance r is connected to a variable external resistor R. When R = 2.0 Ω the current is 2.0 A; when R = 5.0 Ω the current is 1.0 A. **[8]**

1. Write two equations and solve them for E and r.
2. Calculate the current that would flow if the battery terminals were short-circuited.

> **Answer.** **(a)** E = I(R + r): 2.0(2.0 + r) = 4.0 + 2.0r; 1.0(5.0 + r) = 5.0 + r. Equating: 4.0 + 2.0r = 5.0 + r ⇒ r = **1.0 Ω**. Then E = 5.0 + 1.0 = **6.0 V**.
> **(b)** Short circuit (R = 0): I = E/r = 6.0/1.0 = **6.0 A**.

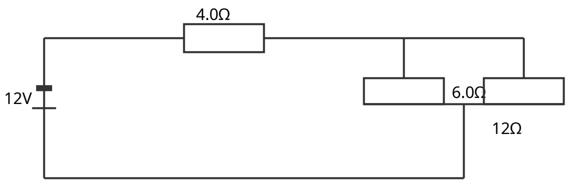
Q121. Distinguish between the electromotive force (e.m.f.) of a source and the potential difference (p.d.) across its terminals. **[6]**

1. Define each in terms of energy per unit charge.
2. Explain why the terminal p.d. is always less than the e.m.f. when the source supplies current, and equal to it when no current flows.

> **Answer. (a)** E.m.f. is the electrical energy transferred per unit charge by the source in driving charge around the whole circuit; terminal p.d. is the energy transferred per unit charge to the external components.
> **(b)** When current I flows, energy Ir per unit charge is dissipated in the internal resistance, so terminal p.d. = E - Ir < E. When I = 0 there are no 'lost volts' (Ir = 0), so the terminal p.d. equals the e.m.f.

Q122. In the circuit shown, a 12 V battery of negligible internal resistance is connected to a 4.0 Ω resistor in series with a parallel combination of a 6.0 Ω and a 12 Ω resistor. **[8]**

1. Calculate the total resistance of the circuit and the current drawn from the battery.
2. Calculate the current in the 6.0 Ω resistor.



> **Answer. (a)** Parallel pair: 1/R = 1/6.0 + 1/12 = 0.25 ⇒ R = 4.0 Ω. Total = 4.0 + 4.0 = 8.0 Ω. Current from battery I = 12/8.0 = **1.5 A**.
> **(b)** P.d. across the parallel pair = 1.5 × 4.0 = 6.0 V. Current in 6.0 Ω = 6.0/6.0 = **1.0 A** (the other 0.5 A is in the 12 Ω).

Q123. A potential divider consists of two fixed resistors, R_1 = 3.0 kΩ and R_2 = 9.0 kΩ, connected in series across a 6.0 V supply. The output is taken across R_2. **[8]**

1. Calculate the output voltage.
2. A voltmeter of resistance 18 kΩ is connected across the output. Calculate the new output voltage and comment.

> **Answer. (a)** V_{out} = V × R_2/(R_1 + R_2) = 6.0 × 9.0/12.0 = **4.5 V**.
> **(b)** R_2 ▯ 18 kΩ = (9.0 × 18)/(9.0 + 18) = 162/27 = 6.0 kΩ. V_{out} = 6.0 × 6.0/(3.0 + 6.0) = **4.0 V**. The voltmeter draws current and 'loads' the divider, reducing the output below the ideal 4.5 V.

Q124. Explain the principle of a potentiometer used to compare two e.m.f.s, and describe the role of the galvanometer. **[7]**

1. Explain how a balance point is found and what it represents.
2. A cell balances at 55.0 cm and a second cell at 43.0 cm on the same uniform wire. If the first cell has e.m.f. 1.50 V, find the e.m.f. of the second.

> **Answer. (a)** A steady current flows through a uniform wire from a driver cell, giving a p.d. proportional to length. The cell under test is connected (through a galvanometer) to one end and a sliding contact; the contact is moved until the galvanometer reads zero (balance). At balance the wire's p.d. over that length equals the cell's e.m.f. and no current is drawn from the cell.
>
> **(b)** E $^{\in}$ length: E_2 = 1.50 × 43.0/55.0 = **1.17 V**.

Q125. A light-dependent resistor (LDR) is connected in series with a 1.5 kΩ fixed resistor across a 9.0 V supply. The output is taken across the fixed resistor and fed to a circuit that switches on a lamp when the output exceeds 6.0 V. **[7]**

1. In darkness the LDR has a resistance of 9.0 kΩ. Calculate the output voltage and state whether the lamp is on.
2. Explain what happens to the output as it becomes brighter, and hence why this arrangement switches the lamp on at night.

> **Answer. (a)** V_{out} = 9.0 × 1.5/(9.0 + 1.5) = 9.0 × 1.5/10.5 = **1.3 V**. Since 1.3 V < 6.0 V, the lamp is **off** in darkness... which is the wrong way round for a night-light, so the output would instead be taken across the LDR (see (b)).
> **(b)** As it gets brighter the LDR resistance falls, so the p.d. across the fixed resistor rises (and that across the LDR falls). To switch a lamp on at night, the output is taken across the LDR: in darkness the LDR resistance is high, giving a high output that exceeds the threshold and turns the lamp on.

Q126. A network has a 6.0 V cell (negligible internal resistance) connected across two parallel branches: branch 1 is a single 10 Ω resistor; branch 2 is a 4.0 Ω and a 2.0 Ω resistor in series. **[7]**

1. Calculate the current in each branch.
2. Calculate the total current drawn from the cell and the total power delivered.

> **Answer. (a)** Branch 1: I = 6.0/10 = **0.60 A**. Branch 2: R = 4.0 + 2.0 = 6.0 Ω, I = 6.0/6.0 = **1.0 A**.
> **(b)** Total current = 0.60 + 1.0 = **1.6 A**. Power = VI = 6.0 × 1.6 = **9.6 W**.

Q127. Two cells are connected in series with each other and with a resistor. Cell A has e.m.f. 1.5 V and internal resistance 0.20 Ω; cell B has e.m.f. 1.5 V and internal resistance 0.30 Ω. They drive a current through an external 2.5 Ω resistor. **[7]**

1. Calculate the current in the circuit.
2. Calculate the terminal p.d. across cell B.

Answer. **(a)** Total e.m.f. = 1.5 + 1.5 = 3.0 V. Total resistance = 2.5 + 0.20 + 0.30 = 3.0 Ω. I = 3.0/3.0 = **1.0 A.**
(b) Terminal p.d. of B = $\varepsilon_B - Ir_B$ = 1.5 - 1.0 × 0.30 = **1.2 V.**

Q128. Using Kirchhoff's laws, derive the formula for the combined resistance of two resistors R_1 and R_2 connected in parallel. **[7]**

1. Give the derivation, stating which law is used at each step.
2. Hence find the single resistance equivalent to 8.0 Ω and 8.0 Ω in parallel.

Answer. **(a)** Both resistors have the same p.d. V (parallel). By Kirchhoff's first law the total current splits: $I = I_1 + I_2$. Using V = IR for each: $V/R = V/R_1 + V/R_2$. Dividing by V: $\mathbf{1/R = 1/R_1 + 1/R_2}$.

(b) 1/R = 1/8.0 + 1/8.0 = 1/4.0 ⇒ R = **4.0 Ω.**

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11 Particle physics

Q129. In the α-particle scattering experiment, a beam of α-particles is directed at a thin gold foil. Most pass straight through, a few are deflected through large angles, and a very small number bounce almost straight back. **[7]**

1. State what each of these three observations tells us about the atom.
2. Explain why the nucleus must be both very small and positively charged.

Answer. **(a)** Most pass straight through ⇒ the atom is mostly empty space. A few deflected through large angles ⇒ there is a concentrated region of positive charge that repels them. A very few bounce back ⇒ this region (the nucleus) is very small and contains most of the atom's mass.
(b) Only a tiny fraction rebound, so the charged core occupies a very small volume; the α-particles (positive) are repelled, so the core must be positively charged.

Q130. A nuclide is represented as ${}^{27}_{13}Al$. **[6]**

1. State the number of protons, neutrons and electrons in a neutral atom of this nuclide.
2. Explain what is meant by an isotope, and give the notation for an aluminium isotope with two more neutrons.

Answer. **(a)** Protons = 13; neutrons = 27 - 13 = 14; electrons (neutral atom) = 13.
(b) Isotopes are nuclei of the same element (same proton number) with different numbers of neutrons (different nucleon number). Two more neutrons gives nucleon number 29: ${}^{29}_{13}\mathbf{Al}$.

Q131. Describe the composition, relative mass and relative charge of α-, β^-- and γ-radiations. **[7]**

1. Give the nature, mass and charge of each.
2. State which is the most ionising and which is the most penetrating, with a brief reason.

Answer. **(a)** α: a helium nucleus (2 protons + 2 neutrons), relative mass 4, charge +2e. β^-: a fast electron, relative mass ≈ 1/1840, charge -e. γ: a high-energy electromagnetic photon, zero mass, zero charge.
(b) α is the most ionising (large charge and mass, many collisions over a short range); γ is the most penetrating (no charge, interacts least, so it travels furthest).

Q132. A nucleus of uranium $^{238}_{92}U$ decays by emitting an α-particle. **[6]**

1. Write the nuclear equation, giving the nucleon and proton numbers of the daughter nucleus.
2. Explain how nucleon number and charge are conserved in this decay.

> **Answer.** **(a)** $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}\alpha$. Daughter (thorium): **nucleon number 234, proton number 90**.
> **(b)** Nucleon number: 238 = 234 + 4 □. Charge (proton number): 92 = 90 + 2 □. Both totals are unchanged before and after the decay.

Q133. During β^- decay a neutron in the nucleus changes into a proton. **[6]**

1. Write the equation for this change at the level of the nucleon, including the emitted particles.
2. Explain why the emitted β-particles have a continuous range of energies.

> **Answer.** **(a)** $n \rightarrow p + {}^{0}_{-1}e + \bar{\nu}_e$ (a proton, an electron and an electron antineutrino).
> **(b)** The available energy is shared between the β-particle and the (anti)neutrino in varying proportions; because the antineutrino carries away a variable amount, the β-particle can have any energy from near zero up to a maximum — a continuous spectrum.

Q134. State the quark composition of a proton and of a neutron. **[7]**

1. Give the compositions and verify the charge of each using quark charges (u = +⅔e, d = -⅓e).
2. State the difference between a baryon and a meson.

> **Answer.** **(a)** Proton = uud: (+⅔) + (+⅔) + (-⅓) = **+1e** □. Neutron = udd: (+⅔) + (-⅓) + (-⅓) = **0** □.
> **(b)** A baryon is made of three quarks (e.g. proton, neutron); a meson is made of one quark and one antiquark.

Q135. During β^+ decay a proton changes into a neutron. **[6]**

1. Describe this change in terms of quarks.
2. State the two particles emitted and identify them as leptons.

> **Answer.** **(a)** A proton (uud) becomes a neutron (udd): one **up quark changes to a down quark**.
> **(b)** A **positron** (e^+) and an **electron neutrino** (ν_e) are emitted; both are leptons (fundamental particles).

Q136. Explain what is meant by an antiparticle, and describe the positron. **[5]**

1. Define antiparticle and state the properties of the positron relative to the electron.
2. State the class of fundamental particle to which the electron and neutrino belong.

Answer. **(a)** An antiparticle has the same mass as its corresponding particle but opposite charge (and opposite other 'charges'). The positron is the antiparticle of the electron: same mass, but charge +e (instead of -e).
(b) They are **leptons**.

Q137. A radium nucleus $^{226}_{88}Ra$ decays by α-emission to radon (Rn), which then decays by α-emission to polonium (Po). **[6]**

1. Write both decay equations, giving the nucleon and proton numbers at each stage.

Answer. First decay: $^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}\alpha$ (radon: **A = 222, Z = 86**).

Second decay: $^{222}_{86}Rn \rightarrow {}^{218}_{84}Po + {}^{4}_{2}\alpha$ (polonium: **A = 218, Z = 84**). (Each α-emission lowers A by 4 and Z by 2.)

Q138. The six flavours of quark are up, down, strange, charm, top and bottom. **[6]**

1. State the charge of the up, down and strange quarks, and of their antiquarks.
2. A particle has quark composition uū (an up quark and an up antiquark). State whether it is a baryon or a meson, and calculate its charge.

Answer. **(a)** u: +⅔e (ū: -⅔e); d: -⅓e (d̄: +⅓e); s: -⅓e (s̄: +⅓e).

(b) One quark + one antiquark ⇒ a **meson**. Charge = (+⅔e) + (-⅔e) = **0**.

Q139. The unified atomic mass unit (u) is used as a unit of mass in nuclear physics. **[5]**

1. State the approximate value of 1 u in kilograms and explain why it is a convenient unit.
2. A carbon-12 atom has a mass of exactly 12 u. Calculate its mass in kilograms.

Answer. **(a)** 1 u = 1.66×10^{-27} kg (one twelfth of the mass of a carbon-12 atom). It is convenient because nucleon masses are then close to 1 u, so nuclide masses are near their nucleon numbers.
(b) Mass = 12 × 1.66×10^{-27} = $\mathbf{1.99\times10^{-26}}$ **kg**.

Q140. Explain why α-particles are emitted with discrete (specific) energies, whereas β-particles from a given decay have a continuous range of energies. **[6]**

1. Explain the α case.
2. Explain the β case, naming the additional particle involved.

> **Answer. (a)** In α-decay the energy released is shared between just two bodies (the α-particle and the recoiling daughter nucleus). Conservation of momentum and energy then fixes the α-particle's energy at one (or a few discrete) value(s) for a given transition.
> **(b)** In β-decay the energy is shared among three bodies, because a (anti)neutrino is also emitted. The neutrino carries a variable share, so the β-particle's energy varies continuously from near zero up to a maximum.

SECTION B — Multiple choice answers [Q1–Q165]

1 Physical quantities and units

1. Which of the following is a set of SI base units only?

- **A** kg, N, s, A
- **B** **m, kg, s, A**
- **C** m, kg, s, J
- **D** m, g, s, A

> **Correct: B.** The SI base units include metre, kilogram, second and ampere. N and J are derived; g (gram) is not the base unit of mass (kg is).

---

2. The unit of the joule expressed in SI base units is:

- **A** kg m s^{-2}
- **B** **kg m^2 s^{-2}**
- **C** kg m^2 s^{-3}
- **D** kg m^{-1} s^{-2}

> **Correct: B.** J = N m = (kg m s^{-2})(m) = kg m^2 s^{-2}.

---

3. A frequency of 2.5 GHz is equal to:

- **A** 2.5×10^6 Hz
- **B** **2.5×10^9 Hz**
- **C** 2.5×10^{12} Hz
- **D** 2.5×10^{-9} Hz

> **Correct: B.** The prefix giga (G) = 10^9, so 2.5 GHz = 2.5×10^9 Hz.

---

4. Which quantity is a vector?

A kinetic energy

B temperature

C **momentum**

D electric charge

Correct: C. Momentum (mass × velocity) has direction. The others are scalars.

---

5. The diameter of a wire is measured as (0.50 ± 0.01) mm. The percentage uncertainty in its cross-sectional area is:

A 2%

B **4%**

C 1%

D 8%

Correct: B. Area $^{\in}$ d^2, so % uncertainty = 2 × (0.01/0.50) = 2 × 2% = 4%.

---

6. Two forces of 3.0 N and 4.0 N act at right angles at a point. The magnitude of their resultant is:

A 1.0 N

B **5.0 N**

C 7.0 N

D 12 N

Correct: B. Perpendicular: $R = \sqrt{3.0^2 + 4.0^2} = \sqrt{25} = 5.0$ N.

---

7. The base units kg m^{-1} s^{-2} are the units of:

A force

B energy

C **pressure**

D power

Correct: C. Pa = N m^{-2} = kg m^{-1} s^{-2}. This is pressure (also stress).

---

8. A physical quantity is estimated. Which is the best estimate of the wavelength of red light?

A 7×10^{-9} m

B **7×10^{-7} m**

C 7×10^{-5} m

D 7×10^{-3} m

Correct: B. Visible light is 400–700 nm; red is at the long-wavelength end, ~7×10^{-7} m.

---

9. A force of 20 N acts at 60° to the horizontal. Its horizontal component is:

- **A** **10 N**
- **B** 17 N
- **C** 20 N
- **D** 12 N

> **Correct: A.** Horizontal component = 20 cos60° = 20 × 0.50 = 10 N.

---

10. A micrometer has a zero error of +0.02 mm. A reading of 3.46 mm is taken. The true measurement is:

- **A** 3.48 mm
- **B** **3.44 mm**
- **C** 3.46 mm
- **D** 3.42 mm

> **Correct: B.** A positive zero error means readings are too high by 0.02 mm, so subtract: 3.46 - 0.02 = 3.44 mm.

---

11. Which of the following pairs contains only scalar quantities?

- **A** **speed and energy**
- **B** velocity and force
- **C** displacement and mass
- **D** acceleration and power

> **Correct: A.** Speed and energy are both scalars. The other pairs each contain at least one vector.

---

12. The percentage uncertainties in P, Q and R are 2%, 3% and 1%. For X = PQ/R, the percentage uncertainty in X is:

- **A** 0%
- **B** 2%
- **C** **6%**
- **D** 5%

> **Correct: C.** For products and quotients, percentage uncertainties add: 2 + 3 + 1 = 6%.

---

13. A measurement is described as accurate but not precise. This means the results are:

- **A** all very close together but far from the true value
- **B** **scattered but their average is close to the true value**
- **C** all identical and correct
- **D** affected by a large zero error

Correct: B. Accurate = mean close to true value; not precise = large random scatter about that mean.

---

14. Which equation is homogeneous with respect to units? (s = displacement, u = initial velocity, a = acceleration, t = time)

- **A** $s = ut^2 + at$
- **B** $s = ut + ½at^2$
- **C** $s = u + ½at^2$
- **D** $s = u^2t + at$

Correct: B. Only $s = ut + ½at^2$ is dimensionally consistent: ut has units m, and $½at^2 = (m\ s^{-2})(s^2) = m$.

---

15. Two forces each of magnitude 6.0 N act at a point with an angle of 120° between them. The magnitude of the resultant is:

- **A** 0 N
- **B** **6.0 N**
- **C** 10.4 N
- **D** 12 N

Correct: B. $R^2 = 6^2 + 6^2 + 2(6)(6)\cos 120° = 36 + 36 + 72(-0.5) = 36 \Rightarrow R = 6.0$ N.

---

2 Kinematics

16. A car travels 30 m north then 40 m east. Its displacement from the start is:

A 70 m

B **50 m**

C 10 m

D 35 m

Correct: B. Displacement = $\sqrt{30^2 + 40^2} = \sqrt{2500}$ = 50 m (the distance travelled is 70 m).

---

17. An object falls freely from rest. The distance it falls in the first 2.0 s is ($g = 9.8$ m s^{-2}):

A 9.8 m

B **19.6 m**

C 39.2 m

D 4.9 m

Correct: B. $s = ½gt^2 = ½(9.8)(2.0)^2 = ½(9.8)(4.0) = 19.6$ m.

---

18. The gradient of a displacement–time graph gives:

A acceleration

B **velocity**

C distance

D force

Correct: B. Rate of change of displacement with time = velocity.

---

19. A ball is thrown vertically up at 20 m s^{-1}. The time to reach its highest point is ($g = 10$ m s^{-2}):

A 1.0 s

B **2.0 s**

C 4.0 s

D 0.5 s

Correct: B. At the top $v = 0$: $t = (v - u)/(-g) = (0 - 20)/(-10) = 2.0$ s.

---

20. The area under a velocity–time graph represents:

- **A** acceleration
- **B** **displacement**
- **C** speed
- **D** force

Correct: B. Velocity × time = displacement, which equals the area beneath the graph.

---

21. A projectile is launched at 30° to the horizontal at 40 m s^{-1}. Its initial vertical velocity component is:

- **A** **20 m s^{-1}**
- **B** 35 m s^{-1}
- **C** 40 m s^{-1}
- **D** 12 m s^{-1}

Correct: A. u_y = 40 sin30° = 40 × 0.50 = 20 m s^{-1}.

---

22. A stone dropped from a bridge takes 3.0 s to reach the water. The height of the bridge is (g = 9.8 m s^{-2}):

- **A** 29 m
- **B** **44 m**
- **C** 88 m
- **D** 15 m

Correct: B. s = ½gt^2 = ½(9.8)(9.0) = 44 m.

---

23. For a body moving with uniform acceleration, which graph is a straight horizontal line?

- **A** displacement–time
- **B** velocity–time
- **C** **acceleration–time**
- **D** kinetic energy–time

Correct: C. Uniform (constant) acceleration gives a horizontal acceleration–time line; v–t is sloped and s–t is curved.

---

24. A car accelerates uniformly from 5.0 to 15 m s^{-1} in 4.0 s. Its acceleration is:

- **A** **2.5 m s^{-2}**
- **B** 5.0 m s^{-2}
- **C** 10 m s^{-2}
- **D** 1.25 m s^{-2}

Correct: A. a = (v - u)/t = (15 - 5.0)/4.0 = 2.5 m s^{-2}.

---

25. A ball is projected horizontally. Ignoring air resistance, its horizontal component of velocity during flight:

- **A** increases uniformly
- **B** decreases uniformly
- **C** **remains constant**
- **D** first increases then decreases

> **Correct: C.** There is no horizontal force, so the horizontal velocity stays constant; only the vertical component changes.

---

26. A car decelerates from 20 m s^{-1} to rest in 50 m. Its deceleration is:

- **A** 2.0 m s^{-2}
- **B** **4.0 m s^{-2}**
- **C** 8.0 m s^{-2}
- **D** 0.4 m s^{-2}

> **Correct: B.** $v^2 = u^2 - 2as \Rightarrow 0 = 400 - 2a(50) \Rightarrow a = 400/100 = 4.0$ m s^{-2}.

---

27. Two balls are released from the same height, one dropped and one thrown horizontally. Ignoring air resistance, they reach the ground:

- **A** **at the same time**
- **B** the dropped one first
- **C** the thrown one first
- **D** depends on the horizontal speed

> **Correct: A.** Vertical motion is identical (same initial vertical velocity of zero and same g), so they land together.

---

28. The velocity–time graph of an object is a straight line with negative gradient that crosses the time axis. At the crossing point the object:

- **A** **is momentarily at rest**
- **B** has maximum acceleration
- **C** has maximum displacement
- **D** reverses its acceleration

> **Correct: A.** Where $v = 0$ the object is instantaneously at rest (e.g. at the top of a throw); acceleration is constant throughout.

---

29. A body starts from rest with uniform acceleration and travels 5.0 m in the first second. In the first two seconds it travels:

A 10 m

B 15 m

C **20 m**

D 25 m

Correct: C. s $\in$ t^2 from rest: s(2 s) = 5.0 × $(2/1)^2$ = 5.0 × 4 = 20 m.

3 Dynamics

30. A resultant force of 12 N acts on a mass of 3.0 kg. The acceleration produced is:

- **A** **4.0 m s^{-2}**
- **B** 36 m s^{-2}
- **C** 0.25 m s^{-2}
- **D** 9.0 m s^{-2}

Correct: A. a = F/m = 12/3.0 = 4.0 m s^{-2}.

---

31. The linear momentum of a 0.50 kg ball moving at 8.0 m s^{-1} is:

- **A** **4.0 kg m s^{-1}**
- **B** 16 kg m s^{-1}
- **C** 0.0625 kg m s^{-1}
- **D** 8.5 kg m s^{-1}

Correct: A. p = mv = 0.50 × 8.0 = 4.0 kg m s^{-1}.

---

32. Force is defined as the rate of change of:

- **A** velocity
- **B** **momentum**
- **C** kinetic energy
- **D** acceleration

Correct: B. Newton's second law: F = Δp/Δt, force is the rate of change of momentum.

---

33. A 1000 kg car experiences a resultant forward force of 2500 N. Its acceleration is:

- **A** 0.4 m s^{-2}
- **B** **2.5 m s^{-2}**
- **C** 4.0 m s^{-2}
- **D** 25 m s^{-2}

Correct: B. a = F/m = 2500/1000 = 2.5 m s^{-2}.

---

34. An object of mass 2.0 kg moving at 3.0 m s^{-1} collides with and sticks to a stationary 4.0 kg object. Their common velocity is:

- **A** 0.5 m s^{-1}
- **B** **1.0 m s^{-1}**
- **C** 1.5 m s^{-1}
- **D** 2.0 m s^{-1}

> **Correct: B.** Momentum: 2.0×3.0 = (2.0+4.0)v ⇒ v = 6.0/6.0 = 1.0 m s^{-1}.

---

35. A ball hits a wall at 6.0 m s^{-1} and rebounds at 4.0 m s^{-1} along the same line. If its mass is 0.20 kg, the magnitude of its change in momentum is:

- **A** 0.40 kg m s^{-1}
- **B** **2.0 kg m s^{-1}**
- **C** 1.2 kg m s^{-1}
- **D** 0.80 kg m s^{-1}

> **Correct: B.** Δp = m(v - u) = 0.20(-4.0 - 6.0) = -2.0; magnitude 2.0 kg m s^{-1}.

---

36. Which is a correct statement of Newton's third law?

- **A** Every action has a bigger reaction
- **B** Forces always act in pairs on the same object
- **C** **If A exerts a force on B, B exerts an equal and opposite force on A**
- **D** The resultant force on a body equals its mass times acceleration

> **Correct: C.** Newton III: the paired forces are equal, opposite, of the same type, and act on different bodies.

---

37. A skydiver reaches terminal velocity when:

- **A** the drag force is zero
- **B** the weight is zero
- **C** **the drag force equals the weight**
- **D** the acceleration equals g

> **Correct: C.** At terminal velocity the resultant force is zero, so drag = weight and velocity is constant.

---

38. An object explodes into two equal masses that fly apart. Compared with each other, the two pieces have:

- **A** equal momenta in the same direction
- **B** **equal and opposite momenta**
- **C** equal kinetic energies but unequal speeds
- **D** zero total kinetic energy

Correct: B. Initial momentum is zero, so the two momenta are equal and opposite (they sum to zero).

---

39. A constant force of 5.0 N acts on a body for 4.0 s. The impulse delivered is:

- **A** 1.25 N s
- **B** 9.0 N s
- **C** **20 N s**
- **D** 0.80 N s

Correct: C. Impulse = $F\Delta t = 5.0 \times 4.0 = 20$ N s (= change in momentum).

---

40. The weight of an object of mass 5.0 kg on a planet where $g = 3.7$ m s^{-2} is:

- **A** 1.4 N
- **B** 8.7 N
- **C** **18.5 N**
- **D** 49 N

Correct: C. $W = mg = 5.0 \times 3.7 = 18.5$ N.

---

41. In an elastic collision, which quantity is NOT necessarily conserved for an individual object?

- **A** total momentum of the system
- **B** total kinetic energy of the system
- **C** **the momentum of one object**
- **D** total energy of the system

Correct: C. System momentum and (for elastic) kinetic energy are conserved, but an individual object's momentum generally changes in the collision.

---

42. A trolley of mass m moving at speed v collides elastically head-on with an identical stationary trolley. After the collision the moving trolley:

- **A** continues at speed v
- **B** **stops, and the other moves off at v**
- **C** rebounds at speed v
- **D** both move off at v/2

> **Correct: B.** For equal masses in an elastic head-on collision the velocities are exchanged: the first stops, the second moves off at v.

---

43. A jet of water of mass flow rate 2.0 kg s^{-1} hits a wall at 10 m s^{-1} and stops. The force on the wall is:

- **A** 5.0 N
- **B** **20 N**
- **C** 0.20 N
- **D** 100 N

> **Correct: B.** F = (mass/s)×Δv = 2.0 × 10 = 20 N.

---

44. The area under a force–time graph gives:

- **A** work done
- **B** power
- **C** **impulse**
- **D** kinetic energy

> **Correct: C.** □F dt = impulse = change of momentum.

---

45. A body of mass 4.0 kg on a rough surface is pulled by a 20 N force but a 8.0 N friction force opposes it. Its acceleration is:

- **A** 7.0 m s^{-2}
- **B** **3.0 m s^{-2}**
- **C** 5.0 m s^{-2}
- **D** 2.0 m s^{-2}

> **Correct: B.** Resultant = 20 - 8.0 = 12 N; a = 12/4.0 = 3.0 m s^{-2}.

---

4 Forces, density and pressure

46. The moment of a 15 N force acting at a perpendicular distance of 0.40 m from a pivot is:

- **A** **6.0 N m**
- **B** 37.5 N m
- **C** 0.027 N m
- **D** 15.4 N m

> **Correct: A.** Moment = F × d = 15 × 0.40 = 6.0 N m.

---

47. A couple consists of:

- **A** a single force through the centre of mass
- **B** two equal forces in the same direction
- **C** **two equal, opposite, parallel forces not in line**
- **D** two unequal forces at right angles

> **Correct: C.** A couple is a pair of equal, opposite, parallel forces whose lines of action do not coincide, producing rotation only.

---

48. The pressure at a depth of 5.0 m in water (density 1000 kg m^{-3}) due to the water alone is (g = 9.8 m s^{-2}):

- **A** 4.9×10^3 Pa
- **B** **4.9×10^4 Pa**
- **C** 5.0×10^5 Pa
- **D** 9.8×10^3 Pa

> **Correct: B.** $\Delta p = \rho g h = 1000 \times 9.8 \times 5.0 = 4.9 \times 10^4$ Pa.

---

49. The density of an object of mass 240 g and volume 30 cm^3 is:

- **A** 8.0 g cm^{-3}
- **B** 0.125 g cm^{-3}
- **C** 7200 kg m^{-3}
- **D** **both A and its SI equivalent 8000 kg m^{-3}**

> **Correct: D.** $\rho = m/V = 240/30 = 8.0$ g cm^{-3} = 8000 kg m^{-3}.

---

50. A uniform beam is in equilibrium. This requires that:

- **A** the resultant force is zero only
- **B** the resultant moment is zero only
- **C** **both the resultant force and the resultant moment are zero**
- **D** all forces act through one point

Correct: C. Equilibrium requires zero resultant force AND zero resultant torque.

---

51. The upthrust on a body fully immersed in a fluid equals the:

- **A** weight of the body
- **B** **weight of fluid displaced**
- **C** volume of the body
- **D** density of the fluid

Correct: B. Archimedes' principle: upthrust = weight of fluid displaced (= ρgV).

---

52. A 2.0 m uniform plank of weight 60 N is pivoted at its centre. A 40 N weight hangs 0.30 m from the pivot. To balance it, a 30 N weight must hang on the other side at a distance of:

- **A** 0.20 m
- **B** **0.40 m**
- **C** 0.60 m
- **D** 0.90 m

Correct: B. Moments balance: $40 \times 0.30 = 30 \times d \Rightarrow d = 12/30 = 0.40$ m.

---

53. A hydraulic system has pistons of area 4.0 cm^2 and 20 cm^2. A force of 30 N on the small piston produces a force on the large piston of:

- **A** 6.0 N
- **B** 30 N
- **C** **150 N**
- **D** 600 N

Correct: C. Equal pressure: $F_2 = F_1(A_2/A_1) = 30 \times 20/4.0 = 150$ N.

---

54. An object floats with 80% of its volume submerged in water (1000 kg m^{-3}). Its density is:

- **A** 200 kg m^{-3}
- **B** **800 kg m^{-3}**
- **C** 1000 kg m^{-3}
- **D** 1250 kg m^{-3}

Correct: B. Fraction submerged = $\rho_{object}/\rho_{fluid} \Rightarrow \rho = 0.80 \times 1000 = 800$ kg m^{-3}.

---

55. The centre of gravity of a body is the point where:

- **A** the mass is greatest
- **B** **all the weight may be taken to act**
- **C** the density is uniform
- **D** the body is most stable

> **Correct: B.** The centre of gravity is the single point at which the entire weight can be considered to act.

---

56. The torque of a couple made of two 12 N forces acting 0.25 m apart is:

- **A** 1.5 N m
- **B** **3.0 N m**
- **C** 6.0 N m
- **D** 48 N m

> **Correct: B.** Torque of a couple = F × separation = 12 × 0.25 = 3.0 N m.

---

57. The SI base units of density are:

- **A** **kg m^{-3}**
- **B** kg m^3
- **C** g cm^{-3}
- **D** kg m^{-2}

> **Correct: A.** Density = mass/volume = kg m^{-3} in SI base units.

---

58. A manometer shows a height difference of 8.0 cm of mercury (density 1.36×10^4 kg m^{-3}). The pressure difference is about:

- **A** **1.1×10^4 Pa**
- **B** 1.1×10^3 Pa
- **C** 1.1×10^5 Pa
- **D** 8.0×10^3 Pa

> **Correct: A.** $\Delta p = \rho g h = 1.36\times10^4 \times 9.81 \times 0.080 = 1.07\times10^4$ Pa.

---

59. A ball of weight 6.0 N is held in water by a string; the upthrust is 2.5 N. The tension in the string is:

- **A** 8.5 N
- **B** **3.5 N**
- **C** 6.0 N
- **D** 2.5 N

> **Correct: B.** Equilibrium: tension + upthrust = weight $\Rightarrow$ T = 6.0 - 2.5 = 3.5 N.

---

5 Work, energy and power

60. A force of 25 N moves an object 4.0 m in the direction of the force. The work done is:

- **A** 6.25 J
- **B** 29 J
- **C** **100 J**
- **D** 21 J

Correct: C. $W = Fs = 25 \times 4.0 = 100$ J.

---

61. A 2.0 kg object moves at 6.0 m s^{-1}. Its kinetic energy is:

- **A** 12 J
- **B** **36 J**
- **C** 72 J
- **D** 6.0 J

Correct: B. $E_K = ½mv^2 = ½(2.0)(36) = 36$ J.

---

62. Power is defined as:

- **A** force times distance
- **B** **work done per unit time**
- **C** energy times time
- **D** force per unit area

Correct: B. Power = work done (or energy transferred) per unit time.

---

63. A load of mass 3.0 kg is raised 2.0 m. The gain in gravitational PE is (g = 9.8 m s^{-2}):

- **A** 6.0 J
- **B** 29 J
- **C** **59 J**
- **D** 15 J

Correct: C. $\Delta E_P = mg\Delta h = 3.0 \times 9.8 \times 2.0 = 58.8 \approx 59$ J.

---

64. A machine has an efficiency of 40%. If the input energy is 500 J, the useful output is:

- **A** 125 J
- **B** **200 J**
- **C** 300 J
- **D** 1250 J

Correct: B. Useful output = 0.40 × 500 = 200 J.

---

65. A car engine delivers a driving force of 800 N at a constant speed of 25 m s^{-1}. The output power is:

- **A** 32 W
- **B** **20 kW**
- **C** 0.032 kW
- **D** 825 W

Correct: B. P = Fv = 800 × 25 = 20 000 W = 20 kW.

---

66. A ball falls freely from rest. When it has fallen half of its total drop height, its speed is what fraction of its final speed?

- **A** 1/2
- **B** **$1/\sqrt{2}$**
- **C** 1/4
- **D** $\sqrt{2}$

Correct: B. $v \in \sqrt{h}$, so at half the height $v = \sqrt{1/2} \times v_{final} = v_{final}/\sqrt{2}$.

---

67. A 60 W lamp is left on for 5.0 minutes. The energy transferred is:

- **A** 300 J
- **B** **1.8×10^4 J**
- **C** 12 J
- **D** 3.0×10^2 J

Correct: B. E = Pt = 60 × (5.0 × 60) = 60 × 300 = 1.8×10^4 J.

---

68. A pendulum bob is released from a height of 0.20 m. Ignoring resistance, its speed at the lowest point is ($g = 9.8$ m s^{-2}):

- **A** 1.4 m s^{-1}
- **B** **2.0 m s^{-1}**
- **C** 3.9 m s^{-1}
- **D** 4.0 m s^{-1}

> **Correct: B.** $v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 0.20} = \sqrt{3.92} = 1.98 \approx 2.0$ m s^{-1}.

---

69. Which statement about a body raised at constant velocity is correct?

- **A** Its kinetic energy increases
- **B** **Its potential energy increases**
- **C** No work is done on it
- **D** Its total energy is constant

> **Correct: B.** At constant velocity KE is unchanged; the work done against gravity increases the gravitational PE.

---

70. The efficiency of a system is defined as:

- **A** total energy input / useful output
- **B** **useful energy output / total energy input**
- **C** power output × time
- **D** energy output - energy input

> **Correct: B.** Efficiency = useful energy (or power) output ÷ total energy (or power) input.

---

71. A crane lifts a 500 kg load at a steady 0.40 m s^{-1}. The useful power is ($g = 9.8$ m s^{-2}):

- **A** 200 W
- **B** 980 W
- **C** **1960 W**
- **D** 1250 W

> **Correct: C.** $P = Fv = mgv = 500 \times 9.8 \times 0.40 = 1960$ W.

---

72. A 0.10 kg ball dropped from 2.0 m rebounds to 1.4 m. The energy lost in the bounce is ($g = 9.8$ m s^{-2}):

- **A** **0.59 J**
- **B** 1.4 J
- **C** 2.0 J
- **D** 0.20 J

> **Correct: A.** Loss = $mg\Delta h = 0.10 \times 9.8 \times (2.0 - 1.4) = 0.10 \times 9.8 \times 0.6 = 0.59$ J.

---

73. The derivation $P = Fv$ applies to a body moving:

- **A** with increasing acceleration
- **B** **at constant velocity with force along the motion**
- **C** in a circle
- **D** against no force

> **Correct: B.** $P = W/t = Fs/t = Fv$ when the force acts along the direction of motion (constant v assumed for the simple derivation).

---

74. A spring stores 8.0 J of elastic PE when compressed by 0.10 m. Its spring constant is:

- **A** 80 N m^{-1}
- **B** 800 N m^{-1}
- **C** **1600 N m^{-1}**
- **D** 160 N m^{-1}

> **Correct: C.** $E = \frac{1}{2}kx^2 \Rightarrow k = 2E/x^2 = 2(8.0)/(0.10)^2 = 16/0.01 = 1600$ N m^{-1}.

---

6 Deformation of solids

75. Hooke's law states that, up to the limit of proportionality, the extension of a spring is:

- **A** inversely proportional to the load
- **B** **proportional to the load**
- **C** proportional to the square of the load
- **D** independent of the load

Correct: B. F = kx: extension is directly proportional to the applied force up to the limit of proportionality.

---

76. A spring extends 5.0 cm under a load of 10 N. Its spring constant is:

- **A** 2.0 N m^{-1}
- **B** 50 N m^{-1}
- **C** **200 N m^{-1}**
- **D** 0.50 N m^{-1}

Correct: C. k = F/x = 10/0.050 = 200 N m^{-1}.

---

77. The Young modulus is defined as:

- **A** stress × strain
- **B** **stress / strain**
- **C** strain / stress
- **D** force / extension

Correct: B. E = stress/strain (the gradient of the linear region of a stress–strain graph).

---

78. Strain has:

- **A** the unit pascal
- **B** the unit metre
- **C** **no unit**
- **D** the unit newton

Correct: C. Strain = extension/original length, a ratio of two lengths, so it has no unit.

---

79. A wire of cross-sectional area 2.0×10^{-7} m² carries a tension of 40 N. The stress is:

- **A** **2.0×10^{8} Pa**
- **B** 8.0×10^{-6} Pa
- **C** 20 Pa
- **D** 2.0×10^{7} Pa

> **Correct: A.** Stress = F/A = $40/2.0\times10^{-7} = 2.0 \times 10^{8}$ Pa.

---

80. The elastic potential energy stored in a spring of constant k extended by x is:

- **A** kx
- **B** ½kx
- **C** **½kx²**
- **D** kx²

> **Correct: C.** $E_P = ½kx^2 = ½Fx$ (area under the force–extension line).

---

81. Beyond the elastic limit, a material:

- **A** obeys Hooke's law
- **B** returns fully to its original length when unloaded
- **C** **does not return to its original length when unloaded**
- **D** has zero stress

> **Correct: C.** Beyond the elastic limit the deformation is plastic (permanent), so the material keeps some extension after unloading.

---

82. The area under a force–extension graph represents:

- **A** the spring constant
- **B** the stress
- **C** **the work done in stretching**
- **D** the strain

> **Correct: C.** Area under F–x = work done (elastic PE stored if within the limit of proportionality).

---

83. A load of 12 N stretches a wire by 0.60 mm. The energy stored (assuming Hooke's law) is:

- **A** 7.2×10^{-3} J
- **B** **3.6×10^{-3} J**
- **C** 7.2 J
- **D** 3.6 J

> **Correct: B.** E = ½Fx = ½ × 12 × $0.60\times10^{-3} = 3.6 \times 10^{-3}$ J.

---

84. Two identical springs of constant k are joined end to end (in series). The combined spring constant is:

- **A** 2k
- **B** k
- **C** **k/2**
- **D** 4k

Correct: C. Series: $1/k_{eff} = 1/k + 1/k = 2/k \Rightarrow k_{eff} = k/2$ (softer, extends more).

---

85. The gradient of the linear part of a stress–strain graph gives the:

- **A** spring constant
- **B** **Young modulus**
- **C** ultimate tensile stress
- **D** elastic limit

Correct: B. Stress/strain in the linear region = the Young modulus.

---

86. A steel wire and a copper wire have the same dimensions and carry the same load. Steel has the larger Young modulus, so the steel wire has the:

- **A** larger extension
- **B** **smaller extension**
- **C** same extension
- **D** larger stress

Correct: B. Same stress (same load and area); extension = (stress/E)L, so the larger E (steel) gives the smaller extension.

---

87. A material that undergoes large plastic deformation before breaking is described as:

- **A** brittle
- **B** **ductile**
- **C** elastic
- **D** stiff

Correct: B. Ductile materials (e.g. copper) can be drawn out with much plastic deformation before fracture; brittle ones break with little plastic deformation.

---

7 Waves

88. A wave has frequency 200 Hz and wavelength 1.5 m. Its speed is:

- **A** 133 m s^{-1}
- **B** **300 m s^{-1}**
- **C** 0.0075 m s^{-1}
- **D** 201.5 m s^{-1}

Correct: B. $v = f\lambda = 200 \times 1.5 = 300$ m s^{-1}.

---

89. The period of a wave of frequency 50 Hz is:

- **A** 50 s
- **B** **0.02 s**
- **C** 0.5 s
- **D** 2.0 s

Correct: B. $T = 1/f = 1/50 = 0.02$ s.

---

90. A sound wave of speed 340 m s^{-1} has a frequency of 680 Hz. Its wavelength is:

- **A** **0.5 m**
- **B** 2.0 m
- **C** 1.0 m
- **D** 0.25 m

Correct: A. $\lambda = v/f = 340/680 = 0.5$ m.

---

91. Which of the following is a longitudinal wave?

- **A** light
- **B** a wave on a string
- **C** **sound in air**
- **D** a water surface wave

Correct: C. In sound the air particles oscillate along the direction of travel — longitudinal. The others are transverse.

---

92. Which region of the electromagnetic spectrum has the longest wavelength?

- **A** ultraviolet
- **B** X-rays
- **C** **radio waves**
- **D** infrared

> **Correct: C.** Radio waves have the longest wavelengths (> 0.1 m) of the listed regions.

---

93. Polarisation can occur only with:

- **A** longitudinal waves
- **B** **transverse waves**
- **C** sound waves
- **D** all waves

> **Correct: B.** Polarisation requires oscillations perpendicular to the direction of travel, which only transverse waves have.

---

94. Plane-polarised light passes through a filter whose axis is at 60° to the polarisation plane. The fraction of intensity transmitted is:

- **A** 0.87
- **B** 0.75
- **C** 0.50
- **D** **0.25**

> **Correct: D.** Malus: $I/I_0 = \cos^2 60° = (0.5)^2 = 0.25$.

---

95. If the amplitude of a progressive wave is doubled, its intensity is:

- **A** halved
- **B** doubled
- **C** **quadrupled**
- **D** unchanged

> **Correct: C.** I $^{\in}$ amplitude^2, so doubling the amplitude multiplies the intensity by 4.

---

96. A source of sound moves towards a stationary observer. Compared with the emitted sound, the observer hears a frequency that is:

A **higher**

B lower

C the same

D zero

Correct: A. For an approaching source the wavefronts bunch up, so the observed frequency is higher (Doppler effect).

---

97. Two points on a progressive wave are separated by a quarter of a wavelength. Their phase difference is:

A 45°

B **90°**

C 180°

D 360°

Correct: B. Phase difference = (¼) × 360° = 90° (π/2 rad).

---

98. In free space, all electromagnetic waves have the same:

A frequency

B wavelength

C **speed**

D amplitude

Correct: C. All EM waves travel at $c = 3.0 \times 10^{8}$ m s^{-1} in free space.

---

99. On a CRO, one complete cycle of a signal spans 5.0 divisions with the time-base at 2.0 ms per division. The frequency is:

A 50 Hz

B **100 Hz**

C 200 Hz

D 500 Hz

Correct: B. $T = 5.0 \times 2.0\times10^{-3} = 1.0\times10^{-2}$ s; $f = 1/T = 100$ Hz.

---

100. The intensity of light from a point source at distance r is I. At distance 3r it is:

- **A** I/3
- **B** I/6
- **C** **I/9**
- **D** 3I

Correct: C. Inverse-square law: $I \in 1/r^2$, so at 3r the intensity is I/9.

---

101. The approximate range of wavelengths of visible light in free space is:

- **A** 40–70 nm
- **B** **400–700 nm**
- **C** 4–7 μm
- **D** 4–7 mm

Correct: B. Visible light spans about 400 nm (violet) to 700 nm (red).

---

102. In a longitudinal wave, the regions where particles are furthest apart are called:

- **A** compressions
- **B** **rarefactions**
- **C** nodes
- **D** crests

Correct: B. Rarefactions are the low-density regions where particles are furthest apart; compressions are high-density.

---

8 Superposition

103. In a double-slit experiment the fringe separation is given by:

- **A** ax/D
- **B** **λD/a**
- **C** λa/D
- **D** aD/λ

Correct: B. x = λD/a, where a is slit separation and D the slit-to-screen distance.

---

104. For a diffraction grating, the maxima occur where:

- **A** a sinθ = nλ
- **B** **d sinθ = nλ**
- **C** d cosθ = nλ
- **D** d sinθ = λ/n

Correct: B. d sinθ = nλ, where d is the grating spacing and n the order.

---

105. Adjacent nodes on a stationary wave are separated by:

- **A** λ
- **B** **λ/2**
- **C** λ/4
- **D** 2λ

Correct: B. Nodes are half a wavelength apart (as are adjacent antinodes).

---

106. Two sources are coherent if they have:

- **A** the same amplitude only
- **B** **a constant phase difference**
- **C** different frequencies
- **D** the same colour only

Correct: B. Coherence means a constant phase difference (and the same frequency).

---

107. Constructive interference occurs when the path difference is:

- **A** an odd number of half-wavelengths
- **B** **a whole number of wavelengths**
- **C** a quarter of a wavelength
- **D** zero only

Correct: B. Path difference = $n\lambda$ gives waves in phase ⇒ constructive interference.

---

108. A stationary wave, compared with a progressive wave, transfers:

- **A** more energy
- **B** **no net energy along the wave**
- **C** energy faster
- **D** energy only at nodes

Correct: B. A stationary wave stores energy and transfers none along its length (energy sloshes between KE and PE).

---

109. A grating has 400 lines per mm. Its slit spacing is:

- **A** **2.5×10^{-6} m**
- **B** 4.0×10^{-4} m
- **C** 2.5×10^{-3} m
- **D** 4.0×10^{5} m

Correct: A. $d = 1/(400\times10^{3}$ per m$) = 2.5 \times 10^{-6}$ m.

---

110. Which condition is essential to observe stable two-source interference fringes?

- **A** sources of different frequency
- **B** **coherent sources**
- **C** sources very far apart
- **D** white light only

Correct: B. The sources must be coherent (constant phase difference) for a stable, observable pattern.

---

111. In a double-slit experiment, moving the screen further away (increasing D) makes the fringes:

- **A** closer together
- **B** **further apart**
- **C** unchanged
- **D** disappear

Correct: B. $x = \lambda D/a \in D$, so increasing D increases the fringe separation.

---

112. At an antinode of a stationary wave, the amplitude of oscillation is:

A zero

B **a maximum**

C half the maximum

D equal to that at a node

Correct: B. Antinodes are points of maximum amplitude; nodes have zero amplitude.

---

113. Light of wavelength 5.0×10^{-7} m strikes a grating of spacing 2.0×10^{-6} m. The angle of the first-order maximum is:

A 7.2°

B **14.5°**

C 30°

D 45°

Correct: B. $\sin\theta = \lambda/d = 5.0\times10^{-7}/2.0\times10^{-6} = 0.25 \Rightarrow \theta = 14.5°$.

---

114. A string fixed at both ends vibrates in its fundamental mode. The wavelength equals:

A L

B **2L**

C L/2

D 4L

Correct: B. The fundamental fits half a wavelength between the ends: $L = \lambda/2$, so $\lambda = 2L$.

---

115. Light of wavelength 600 nm strikes a grating of spacing 2.0×10^{-6} m. The highest order observable is:

A 2

B **3**

C 4

D 5

Correct: B. $n \leq d/\lambda = 2.0\times10^{-6}/6.0\times10^{-7} = 3.3 \Rightarrow n_{max} = 3$.

---

116. Destructive interference between two coherent waves occurs when the path difference is:

- **A** $n\lambda$
- **B** **$(n + \frac{1}{2})\lambda$**
- **C** $\lambda/4$
- **D** $2n\lambda$

Correct: B. An odd number of half-wavelengths, $(n + \frac{1}{2})\lambda$, gives antiphase $\Rightarrow$ destructive interference.

9 Electricity

117. A current of 3.0 A flows for 20 s. The charge transferred is:

- **A** 0.15 C
- **B** 6.7 C
- **C** **60 C**
- **D** 23 C

Correct: C. $Q = It = 3.0 \times 20 = 60$ C.

---

118. The current in a conductor is given by $I = Anvq$. Here v represents the:

- **A** speed of light
- **B** **drift velocity of charge carriers**
- **C** wave speed
- **D** potential difference

Correct: B. v is the mean drift velocity of the charge carriers.

---

119. The potential difference across a component is defined as the:

- **A** charge per unit energy
- **B** **energy transferred per unit charge**
- **C** power per unit current
- **D** current per unit resistance

Correct: B. $V = W/Q$: energy transferred per unit charge.

---

120. A 12 V supply drives a current of 0.50 A through a lamp. The power dissipated is:

- **A** **6.0 W**
- **B** 24 W
- **C** 0.042 W
- **D** 12.5 W

Correct: A. $P = VI = 12 \times 0.50 = 6.0$ W.

---

121. The resistance of a uniform wire is given by $R = \rho L/A$. Doubling the length and doubling the area changes R by a factor of:

- **A** 4
- **B** 2
- **C** **1 (unchanged)**
- **D** ½

Correct: C. $R \in L/A$; doubling both L and A leaves L/A unchanged, so R is unchanged.

---

122. The SI unit of resistivity is:

- **A** Ω
- **B** **Ω m**
- **C** $\Omega\ m^{-1}$
- **D** $\Omega\ m^2$

Correct: B. From $R = \rho L/A$, $\rho = RA/L$ has units $\Omega \times m^2 / m = \Omega\ m$.

---

123. As the current through a filament lamp increases, its resistance:

- **A** decreases
- **B** **increases**
- **C** stays constant
- **D** becomes zero

Correct: B. More current heats the filament; the higher temperature increases the resistance.

---

124. The resistance of a negative-temperature-coefficient thermistor, as temperature rises:

- **A** increases
- **B** **decreases**
- **C** stays constant
- **D** first rises then falls

Correct: B. For an NTC thermistor, resistance decreases as temperature increases.

---

125. The resistance of an LDR when the light intensity falling on it increases:

- **A** increases
- **B** **decreases**
- **C** stays constant
- **D** becomes infinite

Correct: B. More light frees more charge carriers, so the LDR's resistance decreases.

---

126. For the same current, a thinner wire (smaller area) of the same material has a drift velocity that is:

A smaller

B **larger**

C the same

D zero

> **Correct: B.** $v = I/(nAq)$; for fixed I, smaller A gives a larger drift velocity.

---

127. A current of 2.0 A flows through a 5.0 Ω resistor. The power dissipated is:

A 10 W

B **20 W**

C 2.5 W

D 40 W

> **Correct: B.** $P = I^2R = (2.0)^2(5.0) = 20$ W.

---

128. A p.d. of 6.0 V is applied across a 3.0 Ω resistor. The power dissipated is:

A 2.0 W

B **12 W**

C 18 W

D 1.5 W

> **Correct: B.** $P = V^2/R = 6.0^2/3.0 = 36/3.0 = 12$ W.

---

129. Ohm's law applies to a metallic conductor provided that:

A the current is very large

B **the temperature is constant**

C the p.d. is very small

D it is connected in parallel

> **Correct: B.** Ohm's law ($I \in V$) holds only if physical conditions, especially temperature, are constant.

---

130. Electric current is best defined as the:

A number of electrons in a wire

B **rate of flow of charge**

C energy per unit charge

D force on a charge

> **Correct: B.** $I = \Delta Q/\Delta t$: the rate of flow of electric charge.

---

131. That charge is quantised means that charge:

- **A** can take any value
- **B** **exists only in whole multiples of e**
- **C** is always negative
- **D** cannot be measured

Correct: B. Charge comes in integer multiples of the elementary charge e = 1.60×10^{-19} C.

---

132. A charge of 0.32 C passes a point. The number of electrons is (e = 1.6×10^{-19} C):

- **A** **2.0×10^{18}**
- **B** 5.0×10^{18}
- **C** 2.0×10^{19}
- **D** 5.1×10^{-20}

Correct: A. $N = Q/e = 0.32/1.6\times10^{-19} = 2.0 \times 10^{18}$.

---

10 D.C. circuits

133. Three 6.0 Ω resistors in series have a combined resistance of:

- **A** 2.0 Ω
- **B** 6.0 Ω
- **C** **18 Ω**
- **D** 0.5 Ω

Correct: C. Series: R = 6.0 + 6.0 + 6.0 = 18 Ω.

---

134. Two 6.0 Ω resistors in parallel have a combined resistance of:

- **A** 12 Ω
- **B** 6.0 Ω
- **C** **3.0 Ω**
- **D** 1.5 Ω

Correct: C. Parallel of two equal resistors = R/2 = 3.0 Ω.

---

135. A cell of e.m.f. 6.0 V and internal resistance 1.0 Ω is connected to a 2.0 Ω resistor. The current is:

- **A** 3.0 A
- **B** **2.0 A**
- **C** 6.0 A
- **D** 1.0 A

Correct: B. $I = \varepsilon/(R + r) = 6.0/(2.0 + 1.0) = 2.0$ A.

---

136. For the cell in the previous style (E = 6.0 V, r = 1.0 Ω, I = 2.0 A), the terminal p.d. is:

- **A** 6.0 V
- **B** **4.0 V**
- **C** 2.0 V
- **D** 8.0 V

Correct: B. $V = \varepsilon - Ir = 6.0 - 2.0 \times 1.0 = 4.0$ V.

---

137. Kirchhoff's first law is a consequence of the conservation of:

- **A** energy
- **B** **charge**
- **C** momentum
- **D** mass

Correct: B. The junction rule (sum of currents in = sum out) follows from conservation of charge.

---

138. Kirchhoff's second law is a consequence of the conservation of:

- **A** charge
- **B** **energy**
- **C** power
- **D** voltage

Correct: B. The loop rule ($\Sigma E = \Sigma IR$) follows from conservation of energy.

---

139. A potential divider has two equal resistors across a 10 V supply. The output across one resistor is:

- **A** 10 V
- **B** **5.0 V**
- **C** 2.5 V
- **D** 0 V

Correct: B. Equal resistors share the p.d. equally: $V_{out} = 10/2 = 5.0$ V.

---

140. A potentiometer is balanced when the galvanometer reads zero. At balance, the current drawn from the cell under test is:

- **A** maximum
- **B** **zero**
- **C** half the driver current
- **D** equal to the driver current

Correct: B. At balance no current flows through the tested cell, so its true e.m.f. is measured.

---

141. The 'lost volts' in a circuit with internal resistance are equal to:

- **A** IR
- **B** **Ir**
- **C** ε
- **D** V/R

Correct: B. Lost volts = Ir, the p.d. across the internal resistance.

---

142. In a parallel circuit, a 6.0 V supply is connected across a 3.0 Ω resistor. The current in that resistor is:

A 0.5 A

B **2.0 A**

C 18 A

D 9.0 A

Correct: B. I = V/R = 6.0/3.0 = 2.0 A (each parallel branch has the full supply p.d.).

---

143. Two resistors, 4.0 Ω and 8.0 Ω, are in series across 12 V. The p.d. across the 8.0 Ω resistor is:

A 4.0 V

B **8.0 V**

C 6.0 V

D 12 V

Correct: B. I = 12/12 = 1.0 A; V_8 = IR = 1.0 × 8.0 = 8.0 V.

---

144. The e.m.f. of a source is the energy transferred per unit charge:

A to the external resistor only

B **in driving charge round the whole circuit**

C lost in the internal resistance

D when no charge flows

Correct: B. E.m.f. is the total energy per unit charge supplied by the source to drive charge around the complete circuit.

---

145. Two identical resistors R in parallel give a combined resistance of:

A 2R

B R

C **R/2**

D R/4

Correct: C. Two equal resistors in parallel combine to R/2.

---

146. A galvanometer is used in a null method because it:

A measures large currents accurately

B **indicates precisely when the current is zero**

C has a very high resistance

D measures power

Correct: B. In null methods the galvanometer detects the balance point (zero deflection = zero current) very sensitively.

---

147. A cell of e.m.f. 1.5 V has internal resistance 0.30 Ω. The maximum (short-circuit) current is:

- **A** **5.0 A**
- **B** 0.45 A
- **C** 0.20 A
- **D** 4.5 A

Correct: A. $I = \varepsilon/r = 1.5/0.30 = 5.0$ A.

---

148. A 4.0 Ω resistor carries a current of 1.5 A. The power dissipated is:

- **A** 6.0 W
- **B** **9.0 W**
- **C** 2.7 W
- **D** 0.38 W

Correct: B. $P = I^2R = (1.5)^2(4.0) = 2.25 \times 4.0 = 9.0$ W.

---

11 Particle physics

149. In the notation $^{A}_{Z}X$, Z represents the:

- **A** nucleon number
- **B** **proton number**
- **C** neutron number
- **D** mass in u

Correct: B. Z is the proton (atomic) number; A is the nucleon number.

---

150. Isotopes of an element have the same number of:

- **A** neutrons
- **B** **protons**
- **C** nucleons
- **D** electrons in the nucleus

Correct: B. Isotopes have the same proton number but different numbers of neutrons.

---

151. An α-particle consists of:

- **A** **2 protons and 2 neutrons**
- **B** 1 proton and 1 neutron
- **C** 2 electrons
- **D** 2 protons only

Correct: A. An α-particle is a helium nucleus: 2 protons and 2 neutrons (charge +2e, mass 4 u).

---

152. During β^- decay, within the nucleus:

- **A** a proton changes into a neutron
- **B** **a neutron changes into a proton**
- **C** an electron changes into a proton
- **D** a neutron is destroyed

Correct: B. β^- decay: a neutron $\rightarrow$ proton + electron + antineutrino.

---

153. The quark composition of a proton is:

- **A** udd
- **B** **uud**
- **C** uds
- **D** uuu

Correct: B. Proton = uud, giving charge (+⅔ +⅔ -⅓) = +1e.

---

154. The quark composition of a neutron is:

- **A** uud
- **B** **udd**
- **C** uus
- **D** ddd

Correct: B. Neutron = udd, giving charge (+⅔ -⅓ -⅓) = 0.

---

155. When $^{238}_{92}U$ emits an α-particle, the daughter nucleus is:

- **A** $^{234}_{90}Th$
- **B** $^{238}_{90}Th$
- **C** $^{234}_{92}U$
- **D** $^{236}_{91}Pa$

Correct: A. α-emission lowers A by 4 and Z by 2: $^{234}_{90}Th$.

---

156. When a nucleus $^{A}_{Z}X$ emits a β^- particle, the daughter nucleus is:

- **A** $^{A}_{Z+1}Y$
- **B** $^{A}_{Z-1}Y$
- **C** $^{A-1}_{Z}Y$
- **D** $^{A-4}_{Z-2}Y$

Correct: A. β^- decay: Z increases by 1, A unchanged.

---

157. An antiparticle has, compared with its particle, the same mass but:

- **A** the same charge
- **B** **opposite charge**
- **C** no mass
- **D** half the charge

Correct: B. An antiparticle has equal mass and opposite charge (e.g. positron vs electron).

---

158. Which of these is a lepton?

A proton

B neutron

C **electron**

D α-particle

Correct: C. The electron (and neutrino) are leptons — fundamental particles. Protons and neutrons are hadrons.

---

159. A meson is composed of:

A three quarks

B **one quark and one antiquark**

C two quarks

D three antiquarks

Correct: B. A meson = one quark + one antiquark; a baryon = three quarks.

---

160. The charge on an up quark is:

A $+\frac{1}{3}e$

B $\boldsymbol{+\frac{2}{3}e}$

C $-\frac{1}{3}e$

D $-\frac{2}{3}e$

Correct: B. The up quark carries $+\frac{2}{3}e$ (its antiquark carries $-\frac{2}{3}e$).

---

161. In any nuclear decay, which quantities are always conserved?

A mass and charge

B **nucleon number and charge**

C kinetic energy only

D number of protons only

Correct: B. Nucleon number and charge (proton number balance) are conserved in nuclear processes.

---

162. An electron antineutrino is emitted during:

A α-decay

B γ-emission

C **β^- decay**

D β^+ decay

Correct: C. β^- decay emits an (electron) antineutrino; β^+ decay emits a neutrino.

---

163. The α-particle scattering experiment provided evidence that the atom has a:

- **A** uniform distribution of charge
- **B** **small, dense, positive nucleus**
- **C** negative core
- **D** solid structure

Correct: B. The rare large-angle deflections showed a tiny, massive, positively charged nucleus with mostly empty space around it.

---

164. β-particles are emitted with a continuous range of energies because:

- **A** they lose energy in the nucleus
- **B** **a neutrino shares the energy**
- **C** the nucleus recoils fully
- **D** they are not fundamental

Correct: B. The decay energy is shared with an emitted (anti)neutrino, so the β-particle energy varies continuously up to a maximum.

---

165. How many flavours of quark are there?

- **A** three
- **B** four
- **C** **six**
- **D** eight

Correct: C. Six: up, down, strange, charm, top and bottom.
